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The Role of Expectancy in Pavlovian Conditioning

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A review of Pavlovian conditioning in animals and humans reveals a critical role for expectancy in the learning of an association between a conditioned stimulus (CS) and an unconditioned stimulus (US), as well as in the expression of this association in a conditioned response (CR). The automatic and involuntary nature of CRs has traditionally been explained in terms of the formation of excitatory or inhibitory links between representations of the CS and US. However, this view has difficulty accounting for the variety of CRs that are observed, some qualitatively different from those elicited by the US, depending on the imminence of the predicted US and the nature of the CS. Furthermore, in humans, the same anticipatory responses are seen when the CS–US relationship is instructed rather than experienced and when the imminent occurrence of the US is directly instructed, without a mediating CS. These findings suggest an alternative explanation in which CRs are anticipatory responses elicited automatically by a specific state of expectancy of the US. The similarity between Pavlovian conditioning in animals and humans in turn suggests a continuity of core mechanisms for learning and performance. We conclude that research and theory in Pavlovian conditioning should go beyond the search for direct CS–US connections and seek to understand the mechanisms that underlie CS–US contingency knowledge, expectancy states, and the generation of anticipatory responses.

Keywords: expectancy, Pavlovian conditioning, associative learning, prediction error, anticipation

Pavlovian conditioning has come to be viewed as the learning that results from exposure to a predictive relation between two events, one of which, the conditioned stimulus (CS), has little or no initial significance, whereas the other, the unconditioned stimulus (US), has such significance. The consequence of exposure to such a relation is that the CS also becomes significant, eliciting conditioned responses (CRs) that reflect the associated US. For example, a CS that signals food typically elicits approach responses that serve to procure the food and consummatory reflexes that facilitate its ingestion, and one that signals danger elicits defensive responses that serve to avoid or escape the danger and protective reflexes that minimize bodily harm. By providing such objective measures of learning and adaptive behavior, Pavlovian conditioning has had a major impact on both theory and practice in psychology, and it remains an important experimental model in cognitive and behavioral neuroscience.

When Pavlov first developed his conception of conditioning, he used language that implied the learned behavior was anticipatory in nature. For example, he described CSs as signals and referred to language as the "second signal system" (Pavlov, 1927, 1941). Yet the mechanism he proposed to explain conditioning did not involve any role for anticipation or expectancy in the production of the CR. Instead, he adopted the reflexive model of Sechenov (Windholz & Lamal, 1986), postulating that learning consisted in the formation of connections between "centers" in the brain that corresponded to the CS and the US. Specifically, he proposed that excitation passed from the CS center to the US center via the learned stimulus–stimulus (S–S) connection, resulting in the CS coming to elicit CRs similar to those elicited by the US (unconditioned responses [URs]). In this sense, the CS acted as a surrogate or substitute for the US, and Pavlov's position has become known as a stimulus substitution model.

In the century following Pavlov's seminal work, early proposals that conditioning did in fact involve the formation of an expectancy of the US (e.g., Tolman, 1933) had little impact, being seen as mentalistic by the dominant stimulus–response (S–R) framework of the time. Models within this framework (e.g., Guthrie, 1935; Hull, 1943; Spence, 1956) assumed a direct connection between the CS and the UR, such that the US itself was not directly encoded. However, in recent decades, terms like anticipation, expectancy, and prediction have become widely used (e.g., Bolles, 1972; Kamin, 1969; Rescorla & Wagner, 1972; Wagner & Rescorla, 1972). Animal subjects are now commonly said to learn that the CS predicts the occurrence of the US and to expect or anticipate the US in the

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presence of the CS. Indeed, it could be concluded that expectancy theories have become the dominant explanatory model in the field. For example, Kirsch et al. (2004) stated that "there is now virtually universal agreement that conditioning involves the production of expectancies" (p. 371).

However, we suggest that there is a major mismatch between the rhetorical language used to describe contemporary models of conditioning and the mechanisms they propose for encoding and utilizing what has been learned, mechanisms that are surprisingly close to Pavlov's model of S–S reflexive activation. Current models are either silent with respect to the mechanisms that underpin learning and performance (e.g., they are mathematical in format), or they explicitly propose that learning generates excitatory and inhibitory links that serve only to increase or decrease the activation of a US representation.

For example, the highly influential model of Rescorla and Wagner (1972) is often described in expectancy terms. Generations of students have been taught that the equation at the heart of the model represents a prediction error: a discrepancy between what the learner expected to occur (ΣV) and what actually occurred (λ) on a given trial. However, the output of the model is represented in a single bidirectional dimension of associative strength (V) ranging from excitation to inhibition. This dimension is said to indicate the effect of the CS on the activation level of the US representation, essentially the same mechanism as proposed by Pavlov.

Wagner's extensions of the Rescorla–Wagner model (Wagner, 1981; Wagner & Brandon, 1989) go further than most associative models in attempting to define the functional mechanisms underlying learning and performance. These models built on the architecture of early information-processing theories of memory and cognition (Atkinson & Shiffrin, 1968). They distinguish between partial (A2) and full (A1) activation of perceptual nodes that make up a stimulus representation, allowing them to account for phenomena such as blocking, habituation, and latent inhibition. However, these models assume that US nodes are in exactly the same state of activation (A2) when primed associatively by the CS as when the US itself has recently been presented. Thus, like the Rescorla and Wagner's (1972) model and other S–S theories, Wagner's models make no fundamental distinction between expecting and remembering a US (Jozefowiez, 2018).

So, can models based entirely on the activation level of US representations actually capture the notion of expectancy? Kirsch et al. (2004) noted that "Contemporary mechanistic accounts of … conditioning typically involve the hypothesis that direct associations between the stimulus and outcome representations … are formed during conditioning." They went on to say that "Obviously, the outcome representations activated by these associations constitute a low-level form of outcome expectancy" (p. 371). Dickinson (1989) similarly argued that associative links are consistent with a "mechanistic" or "weak" version of expectancy theory. In other words, such links are taken to instantiate at least some aspects of the idea of expectancy. Effectively, they allow researchers to talk about expectancy without commitment to the reality of an expectancy.

In this article, we review the evidence concerning the role of expectancy in Pavlovian conditioning, first in animals and then in humans. We argue that a weak version of expectancy theory based on associative links is inconsistent with a range of empirical findings concerning both learning and performance and that many of these findings can be better understood within a "strong" version that accepts the psychological reality of an expectancy. We outline what a genuine expectancy theory might look like, identify challenges for such a theory, and indicate some implications for neuroscience, clinical applications of conditioning, and future research.

Our working definition of expectancy is minimal, partly to avoid prematurely constraining the concept and partly to acknowledge how little we yet know about it. First, expectancy of the US is a distinct psychological state, different from experiencing, remembering, or imagining the event. Second, it is representational, consisting in a belief that the US will occur in the near future. Third, expectancy has consequences; it is capable of eliciting and modulating behavior, including emotion. We restrict our use of expectancy to refer to a current state of anticipation of a forthcoming US, as opposed to a conditional expectancy that if "X" occurs then the US will follow. The elements of this definition build on conceptual work by Dickinson (1989) but differ by imposing no *a priori* restrictions on how expectancies generate behavior.

Pavlovian Conditioning in Animals

The contemporary analysis of conditioning makes a strong distinction between learning—the acquisition and encoding of knowledge—and performance—the subsequent retrieval and expression of that knowledge in behavior. We consider the role of expectancy in learning and performance separately.

Learning

Learning Is Stronger When the CS Closely Precedes the US in Time

The vast majority of conditioning research has followed Pavlov's forward arrangement, in which the CS precedes the US in time. Pavlov also explored simultaneous and backward arrangements where the CS co-occurs with or follows the US, reporting that they produced far fewer CRs than the forward one. In fact, he found evidence that the backward arrangement could generate inhibitory learning. In the forward arrangement, he observed that CRs were more easily obtained when the CS preceded the US by a short rather than a longer interval. Subsequent research has confirmed this finding in a range of other paradigms, albeit with a wide range of absolute intervals (e.g., Kimble, 1947; Revusky, 1968; Smith et al., 1969; for reviews, see Boakes & Costa, 2014; Gormezano & Kehoe, 1981). Thus, the optimal procedure for conditioning appears to be a forward arrangement in which the CS precedes the US by a relatively short interval (high temporal contiguity). This general finding provides prima facie evidence that excitatory conditioning involves learning a temporal relationship between two stimuli such that the first stimulus (CS) acquires the capacity to elicit responding appropriate to the imminent occurrence of the second (US).

Accordingly, the empirical evidence is consistent with the generalization that CRs are produced by stimuli that predict the arrival of the US in the near future. This "informational" perspective inspired the development of theories based on the predictive value of CSs (e.g., Pearce & Hall, 1980; Rescorla & Wagner, 1972; Wagner & Rescorla, 1972). We discuss the role of expectancy in these theories in the next section. Here, we note that prediction-based theories are compatible with three fundamental principles relevant to the adaptive value of Pavlovian conditioning. The first is temporal precedence in causation—causes precede their effects in time. This alignment has encouraged the view that conditioning allows organisms to learn the causal structure of their environment and to benefit from that learning by anticipating future outcomes and regulating their behavior to procure attractive outcomes and avoid aversive ones (e.g., Dickinson, 1980). The second principle is the pragmatic value of temporal information: stimuli that predict a US allow preparatory action even if they are not causal. For example, the sound of a cracking twig might allow escape from a predator even though it is not a cause of that predator. The informational perspective also suggests that there will be an optimal level of (forward) contiguity—stimuli too far in advance of the US do not allow precise prediction, and those too close to the US will not allow time for preparatory action. The third principle is that stimuli that do not predict a US may nonetheless be encoded as significant if they provide information about the future *absence* of that event. For example, a stimulus that signals the absence of danger allows other adaptive behaviors such as feeding to proceed in safety (Moscovitch & LoLordo, 1968).

The impact of CS–US contiguity on learning per se is not yet fully understood and appears to involve multiple processes (see Boakes & Costa, 2014, for a review). For example, the maximum CS–US interval that supports learning varies substantially over conditioning preparations and response systems, from less than a second in the case of eyeblink conditioning to several hours in the case of taste aversion learning. Longer intervals also allow great interference from intervening stimuli, either through associative competition or by consuming attentional resources (Revusky, 1971). A more complete delineation of the role of contiguity in learning per se may require the use of indirect paradigms such as savings, cue competition, and higher order conditioning in order to measure learning that does not meet performance thresholds. Nonetheless, the critical point for present purposes is that the temporal relationship between two stimuli must be encoded during learning, not just their contiguous association.

Expectancy Violation Drives Learning

The informational perspective encouraged the view that animals learn predictive relationships within their environment (S–S associations) rather than learning how to respond to particular stimuli (S–R associations). Two empirical phenomena in particular suggested that learning is driven by expectancy violation. The first was renewed research into inhibitory learning (e.g., Rescorla, 1969). In Pavlov's conditioned inhibition procedure, CS A is paired with a US on some occasions (A+ trials), but not on other occasions when it is presented in compound with a second CS B (AB− trials). This procedure establishes B as a conditioned inhibitor; it reduces the CR to a separately trained excitor (summation test), and it requires more pairings to elicit a CR when subsequently paired with the US. The critical point here is that it is the *absence* of the US on AB− trials that drives inhibitory learning to B. It is hard to see how the absence of an event can lead to learning unless that event was expected in the first place. For example, AB− trials alone do not establish B as a conditioned inhibitor (Rescorla, 1969). Thus, the A+ trials are critical, and a natural interpretation of the role of these trials is that they provide the expectation of the US that is violated on AB− trials, thus generating inhibitory learning. The same principle can be applied to extinction, where it is the CS itself that provides the expectancy of the US.

The second phenomenon that highlighted the role of expectancy violation was the blocking effect (Kamin, 1969). In this design, one CS, A, is followed by the US. In a second phase, an AB compound is also followed by the US. When tested alone, stimulus B shows little evidence of conditioning, despite having been repeatedly paired with the US, in contrast to a control condition in which A was not pretrained in the first phase. Kamin interpreted this finding as meaning that learning only occurs when the US is surprising, that is, not already predicted by another stimulus. The idea is that the added cue B is informationally redundant and hence is not learned about.

These and other findings inspired models that use prediction error to drive learning. The classic error correction model is the Rescorla–Wagner model (Rescorla & Wagner, 1972; Wagner & Rescorla, 1972). It conceptualizes associative strength as a single bipolar dimension with excitation at one end and inhibition at the other. Increases and decreases in associative strength on each trial are driven by positive and negative discrepancies, respectively, between λ (the outcome of the trial: US or no US) and $\sum V$ (the summed associative strength of all stimuli present), commonly described as the subject's expectancy of the outcome on that trial. Importantly, the use of prediction error to drive compensatory changes in associative strength ensures that error will reduce across experiences, allowing learning to track relationships in the environment and adapt when those relationships change. Error correction models are thus compatible with normative (for example, Bayesian) models of causal learning, which use new information from the environment to update prior beliefs (e.g., Courville et al., 2006).

Finally, it is instructive to note that error correction models can fail to account for learning when they depart from expectancy principles. For example, the Rescorla–Wagner model, by conceptualizing inhibition as symmetrically opposite to excitation, makes the prediction that presentation of an inhibitory CS alone (e.g., exposure to B after A+/AB− conditioned inhibition training) will lead to extinction of its inhibitory properties. Yet in expectancy terms, there is no prediction error here: The expected outcome is no US, and the actual outcome is no US. Several studies have confirmed the expectancy prediction rather than the Rescorla–Wagner prediction (e.g., DeVito & Fowler, 1987; Zimmer-Hart & Rescorla, 1974).

The US Is Encoded During Learning

An expectancy theory—either weak or strong—requires that the CS–US relation is directly encoded, such that subsequent encounters with the CS can elicit an expectancy of the US. Historically, the main alternative was that subjects form a mental link between the CS and the UR; in other words, they learn a S–R association. However, as noted, contemporary theories have adopted an informational approach in which subjects encode a predictive relationship between the CS and US; in other words, they learn a S–S association.

There are three main lines of evidence for S–S associations. The first is sensory preconditioning. Two affectively neutral stimuli (S2 and S1) are paired in Stage 1, and one of these stimuli (S1) is paired with a US in Stage 2. Testing in Stage 3 reveals that S2 elicits CRs appropriate to the US. At the time of the S2–S1 pairings, S1 did not produce CRs, so the animals must have directly learned the S–S relationship (Holmes et al., 2022). Second, Kamin's (1969) blocking effect shows that contiguity between a CS and a UR is insufficient for conditioning to occur if the added CS fails to also provide nonredundant predictive information about the US. Third, when the US is revalued after conditioning has taken place, it leads to an immediate reduction in CRs to the CS that had been paired with it. For example, if fear is established to a CS by pairing it with a startle-eliciting noise and the UR to the noise is subsequently habituated by repeated presentations, fear of the CS is eliminated (Rescorla, 1973). A similar result is seen in appetitive conditioning, where a trained CS fails to elicit CRs when tested after devaluation of the food US (Holland & Straub, 1979). Conversely, an upward revaluation of a US can induce Pavlovian approach to a CS that previously elicited avoidance (Robinson & Berridge, 2013).

Occasional failures of US revaluation to influence CRs have been taken to support the idea that S–R associations provide a separate mechanism for Pavlovian conditioning (e.g., Holland & Rescorla, 1975; Rizley & Rescorla, 1972). However, this interpretation does not necessarily follow. First, failures of revaluation are a null result that could reflect a lack of sensitivity or statistical power. Second, some Pavlovian CRs may derive from a general rather than a specific US expectancy, which would not be contradicted by revaluation of the US. For example, Konorski (1967) suggested that different associations are formed when the CS-US interval is short and consistent as opposed to long and variable (or partially reinforced). In the former case, Konorski argued that the CS becomes associated with a detailed "epicritic" sensory representation of the US that supports specific consummatory or defensive responses, whereas in the latter case it is only associated with a general "protopathic" representation that supports preparatory responses such as arousal or fear. We will come back to this idea in the next section on performance. A related proposal is that the US representation may be "defocused" in such a way that CRs are resistant to US devaluation (Dayan & Berridge, 2014).

Performance

The CR Is Not the Same as the UR

Any model that proposes a direct link between CS and US representations must predict that the CR will have the same character as the UR. It may be weaker or include only a subset of the features of the UR, but it should not be qualitatively different. Conversely, the observation of a qualitative difference between the CR and UR would indicate that anticipation of the US activates a different state from that which is activated by the actual presence of the US and that this state elicits systematically different responses from those elicited by the US. In many cases, the CR does in fact appear to be qualitatively the same as the UR. For example, an acidic US elicits high levels of dilute saliva, as does a CS that predicts such a US (Pavlov, 1927). Similarly, a mild electric shock adjacent to the eye elicits a blink response, as does a CS that immediately precedes that US (Gormezano et al., 1962). However, the situation is more complex when a wider range of response systems is considered. For example, in eyeblink conditioning, the shock US elicits heart rate acceleration, whereas the CS elicits a heart rate deceleration, albeit sometimes followed later in training by heart rate acceleration (McEchron et al., 1992; Powell & Levine-Bryce, 1988).

One way of understanding these conflicting results is in terms of Konorski's (1967) distinction between preparatory responses and consummatory/protective responses. He suggested that preparatory CRs involve emotional or motivational states such as appetitive activation and anxiety/fear, which are acquired quickly and do not require close CS–US contiguity. By contrast, consummatory/protective CRs are specific to the US employed and require more pairings and a short, consistent CS–US interval. As his naming suggests, Konorski saw preparatory CRs as anticipatory in nature, and hence they may deviate from the UR, whereas consummatory/protective CRs are typically the same as the UR. In the eyeblink conditioning example above, eyeblinks elicited by the CS would be labeled a protective CR, and the heart rate acceleration would be labeled a preparatory CR. Wagner's ÆSOP model instantiates a similar distinction between emotional responses controlled by contextual cues or long-duration CSs and discrete responses elicited by shorter duration cues (Gewirtz et al., 1998; Wagner & Brandon, 1989). Similarly, in appetitive conditioning, CSs that are closely and consistently paired with food elicit consummatory CRs similar to the UR (e.g., salivation), whereas long-duration CSs and those with a less specific association with the US, such as the context where conditioning occurs, elicit preparatory CRs (e.g., increased activity) that are quite distinct from the UR (Holland, 1980; Lovibond, 1980).

An even more dramatic example of apparent divergence between CRs and URs needs to be considered at this point. In some procedures, especially those involving interoceptive stimuli such as drug administration, the CR appears to be opposite to the UR. For example, morphine produces analgesia, but a CS for morphine produces hyperalgesia (McNally & Westbrook, 1998; Siegel, 1975). Such findings led to the view that conditioning provides a basis for drug tolerance by recruiting drug-opposing CRs in advance of drug delivery, thereby minimizing the impact of the drug on the body (Siegel, 1975). Such a mechanism would clearly be adaptive, but is it an example of an anticipatory CR differing from the UR? Eikelboom and Stewart (1982) have offered a compelling alternative analysis. They pointed out that opponent responses occur when the drug in question exerts its effects peripherally rather than by a centrally mediated pathway. A clear example is intravenous administration of glucose, which increases blood sugar in a direct fashion, not as a response by the body that could in principle be mimicked by a CS. In such cases, the drug is likely to activate peripheral receptors (the actual US) that in turn trigger a centrally mediated homeostatic response (in the case of glucose, the release of insulin to reduce blood sugar). Eikelboom and Stewart argued that this homeostatic response is the true UR and that a CS paired with glucose administration should elicit the same response, namely hypoglycaemia, which is indeed the case (Woods, 1976). They review evidence from a range of drugs that supports the conclusion that peripherally and centrally mediated drug effects condition indirectly and directly, respectively, but in both cases, the CR matches the UR, defined as a centrally mediated response. This analysis offers a parsimonious account of drug CRs and suggests that the anticipatory response to a drug is in fact the same as that to the drug.

However, other CR–UR differences are not so readily explained away. Perhaps the best documented example of a dissociation between the UR and the corresponding preparatory CR is the case of conditioning with a painful US, such as the electric shock commonly used in fear conditioning. Bolles and Fanselow (1980) reviewed evidence that the US and CS activate distinct systems, pain and fear, respectively. They argued that the pain system serves to deal with injury, promoting recuperative behaviors such as licking and resting. The fear system, by contrast, serves to help prevent pain and injury and promotes protective behaviors such as aggression and avoidance (fight/flight responses). In this conceptualization, shock activates pain, whereas a CS paired with shock activates fear, a component of which actually attenuates the anticipated pain via the release of endogenous opioids (Bolles & Fanselow, 1980; Harris et al., 1995).

S–R models of conditioning sought to explain the difference between the responses elicited by a shock US and a CS for shock by arguing that fear is a component of the UR to a painful US and is susceptible to conditioning, whereas the pain component of the UR, for some unspecified reason, cannot be conditioned. However, Blanchard and Blanchard (1969) demonstrated that postshock freezing (indicative of fear) is attenuated if the subjects are tested in a different context. Bolles and Fanselow (1980) replicated this finding and provided a cogent explanation for it. They pointed out that the attenuation of freezing resulting from a context shift contradicts the idea that fear is part of the UR to shock, since it should have been observed regardless of context. Rather, the data suggest that the fear shown in the shocked context was due to one-trial conditioning to that context. A natural interpretation is that the animals tested in the same context were not fearful simply because they had been shocked; rather, they were fearful because they expected a further shock.

The CR Varies as a Function of the CS–US Interval

Konorski's distinction between preparatory and consummatory/protective responses is but one of several proposals as to how CRs vary with the CS–US interval. For example, Ratner (1967) proposed a hierarchy of "defensive distance" based on the proximity or imminence of a threat such as a predator. Such a hierarchy suggests that the responses triggered by a CS that signals an aversive US will vary as a function of the CS–US interval, depending on the defensive repertoire of the species in question. For example, CSs that signal a distal threat might elicit freezing; CSs for an intermediate threat might elicit escape/avoidance, and CSs for a proximal threat might elicit aggression (see also Fanselow & Lester, 1988; Timberlake, 2001). A similar pattern occurs in appetitive conditioning, where behaviors such as searching, rearing, and magazine approach are differentially expressed across CS–US intervals (Holland, 1980; Silva & Timberlake, 1997).

Expectancy Violation Impacts on Behavior

A variety of contrast effects support the idea that animals form expectancies. The classic experiment by Crespi (1942) showed that rats shifted from a small reward to a larger one responded more (they ran faster) than those trained throughout on the larger reward. Conversely, rats shifted from a large reward to a smaller one ran more slowly than rats trained throughout on the smaller one. An even more dramatic impact of the emotional impact of expectancy violation is the distinct behaviors (e.g., escape) that occur when a predicted reward is omitted; behaviors taken to indicate an aversive state of frustration (Amsel, 1992). These behaviors are not seen to nonrewarded control stimuli that have never predicted reward (Papini & Dudley, 1997). Such effects seem to demand a role for expectancy; how can the absence of an event lead to frustration unless that event was expected in the first place? Amsel's (1992) account of frustrative nonreward assumed a causal role for expectancy, holding that frustration is an UR to the nonoccurrence of an attractive outcome that was expected. For a review of contrast effects, see Flaherty (1982).

Expectancy Has Unique Stimulus Properties

Pavlovian associations embedded in instrumental tasks are thought to modulate instrumental behavior (e.g., Konorski, 1967; Rescorla & Solomon, 1967; Trapold & Overmier, 1972). One of the clearest examples of the stimulus properties of anticipation is the differential outcomes effect (Trapold, 1970): the finding that a conditional discrimination is learned faster when each correct response choice leads to a distinct outcome (e.g., food vs. sucrose) rather than a common outcome. Trapold and Overmier (1972) proposed that anticipation of a particular outcome evokes a distinct state that serves as an additional discriminative stimulus to guide instrumental choice. Although stimulus feedback from a CR could in principle mediate such an effect (e.g., Hull, 1943), it is not clear that overt responses possess the necessary specificity. Trapold and Overmier (1972) argued instead that the critical mediator is a central representation of the forthcoming outcome, which they referred to as an expectancy.

In follow-up research, Peterson et al. (1987) trained pigeons in a delayed conditional discrimination task in which food and no food served as posttrial outcomes and/or as pretrial conditional cues. Thus, expectancies of future food and/or memories of previous food could serve as discriminative cues to guide instrumental response choices. Pigeons that had only food expectancy as a guide for choice performed better than those that only had food memory. Furthermore, pigeons that had both available performed no better than those that only had expectancy, particularly at long delays. These results indicate that memories and expectancies are not substitutable for each other and that expectancies can be sustained over longer delays. Linwick and Overmier (2006) used a similar task but, in addition, interspersed differential Pavlovian autoshaping training with the same food outcomes. When they carried out a Pavlovian to instrumental transfer test, they found the Pavlovian cues controlled the same response choices as those cued by posttrial outcomes. They concluded that "associatively activated representations of food-related events resemble expectancies more closely than they do memories."

These studies directly contradict the idea that CSs produce CRs by activating the same central representation as the US. They also challenge Wagner's (1981) view that the US representation activated by a CS and by a recently presented US are the same: a so-called A2 state of activation. As noted by Jozefowiez (2018), the same mental state cannot represent both an expectancy and a memory. Instead, the evidence supports a simple conclusion: CRs are a product of a distinctive state of expectancy of the US, rather than a nonspecific activation of the US representation or a state analogous to a memory trace of the US.

Pavlovian Conditioning in Humans

Empirically, conditioning in humans is similar to that in animals. Virtually all of the signature phenomena observed in animals, from acquisition and extinction to more complex effects such as second-order conditioning, blocking, conditioned inhibition, and conditional discriminations, are also observed in humans. Such similarity suggests there is continuity of mechanism. At the same time, humans can report what they have learned through language, indicating that they can encode and use associative knowledge in abstract, symbolic terms. Accordingly, a critical question that we need to address before considering the role of expectancy in human Pavlovian conditioning is the relationship between anticipatory CRs and verbalizable associative knowledge. Do humans have one associative learning mechanism or two? The traditional answer to this question has been "two": an automatic, unconscious, "implicit" link formation mechanism presumed to be similar to that of animals, and in addition, a symbolic, conscious, "explicit" cognitive system that is not shared with animals except, perhaps, with primates to a limited extent (e.g., Clark & Squire, 1998). However, there is surprisingly little evidence that uniquely supports this position. For reviews of this evidence, see Boakes (1989), Brewer (1974), Dawson and Schell (1985), Lovibond and Shanks (2002), Mertens and Engelhard (2020), and Mitchell et al. (2009a). The principal empirical findings can be summarized as follows:

First, there is little, if any, reliable evidence for unconscious conditioning in humans, and many claimed examples of unconscious conditioning are not readily replicable or are subject to known artefacts such as insensitive or incomplete assessment of relevant contingency knowledge (e.g., Shanks, 2017; Singh et al., 2013; Wiens et al., 2003; see Shanks & St. John, 1994, for a relevant theoretical framework). By contrast, when assessed appropriately, conscious knowledge of CS–US relationships ("contingency awareness") is strongly related to conditioning, both when assessed between participants and within participants over time. This pattern is seen using a variety of assessments, including skin conductance, eyeblink, and potentiated startle (e.g., Dawson & Biferno, 1973; Lovibond et al., 2011; Purkis & Lipp, 2001; Weidemann & Antees, 2012; Weidemann et al., 2013).

Second, dual-system models predict that cognitive load and other manipulations that divert attention and/or occupy working memory will predominantly impact on explicit verbal measures of learning. In fact, however, such manipulations have been shown to impair the acquisition of *both* CRs and verbalizable knowledge (e.g., Carter et al., 2003; Dawson & Biferno, 1973; Purkis & Lipp, 2001; Weidemann et al., 2016).

Third, similar patterns of CRs are generated by direct experience (conditioning) and by verbal instruction (for a review, see Mertens et al., 2018).

Finally, phenomena such as blocking, retrospective revaluation, and generalization follow principles of deductive and inductive reasoning and align with participants' self-reports of the rules they employed to solve the task (De Houwer et al., 2002; Lee et al., 2018, 2019; Lovibond et al., 2003; Mitchell & Lovibond, 2002).

These findings support an alternative "single system" model of human conditioning (De Houwer, 2009; Mitchell et al., 2009a). According to this model, perceptual and memorial processes are of course engaged by conditioning protocols but are not alone capable of producing CRs; instead, they work cooperatively with symbolic processes such as reasoning to produce learning and hence performance of CRs. The outcome of learning is contingency knowledge, which is available for self-report. A corollary of this position is that conditioning in laboratory animals is achieved by mechanisms that foreshadow human cognition. These mechanisms, specifically those possessed by a common mammalian ancestor, evolved into the human cognitive system. Such a pathway is consistent with the evolutionary principle of gradual change that builds on existing capacities. It allows for both quantitative change as well as qualitative change such as the development of language, but critically, it involves continuity. By contrast, the dual system view is anthropocentric and is more difficult to reconcile with principles of evolution. It suggests that humans inherited an associative module unchanged from earlier mammals and at the same time developed a sophisticated cognitive system with no immediate precedent, resulting in two competing systems to solve the same problem (Mitchell et al., 2009b). For further discussion regarding the dual versus single system debate, see the commentaries published with Mitchell et al. (2009a, pp. 198–229) and the rejoinder by Mitchell et al. (2009b), as well as a defense of the dual system position by McLaren et al. (2014).

Given that self-report can illuminate learning mechanisms in humans, we will include data from self-report measures in our review of the role of expectancy in human conditioning. We will also refer to findings from predictive and causal learning tasks based entirely on self-report rather than anticipatory response measures. In such tasks, participants are exposed to contingencies between antecedent and outcome stimuli that are either arbitrary (e.g., shapes) or symbolic (e.g., verbal descriptions of events in a hypothetical scenario, such as foods and allergic reactions in a medical setting). Learning is assessed by predictive (outcome expectancy) ratings, often collected on every trial, and/or causal or contingency ratings, usually collected at the end of the experiment. These tasks typically yield results that are consistent with those from conditioning tasks in humans and animals (Dickinson & Shanks, 1985). Nonetheless, we will indicate when particular findings have only been documented on self-report measures.

Learning

Learning Is Stronger When the CS Closely Precedes the US in Time

As in animals, forward predictive relationships typically generate stronger excitatory responding than simultaneous or backward relationships in human conditioning procedures (e.g., Klosterhalfen et al., 2000; Málková et al., 1989). That said, simultaneous relationships can produce strong excitatory learning when evaluative or affective measures are employed (Gawronski & Mitchell, 2014; Hofmann et al., 2010). Backward relationships also sometimes produce excitatory learning (Prével et al., 2016; Zeiner & Grings, 1968) but more commonly inhibitory learning (e.g., Andreatta & Pauli, 2017; Luck & Lipp, 2017; Urcelay et al., 2008), although there does not appear to be any demonstration in humans to parallel the animal finding that inhibition is more likely with longer training. There is, however, evidence that excitatory backward conditioning depends on how participants represent the experimental procedure. Zeiner and Grings (1968) found that excitatory backward conditioning was only significant in a subgroup who thought that the CS "should have" cued the US. Backward conditioning is also facilitated by instructions that emphasize the causal potency of the CS (Green et al., 2021). These findings suggest that excitatory backward conditioning may be due to temporal uncertainty and/or the false belief that the CS precedes the US, most likely early in training. However, when the temporal relationships are made clear, humans, like animals, integrate temporal information. For example, Arcediano et al. (2005) exposed subjects (animals and humans) to forward parings of two stimuli, S2 followed after a brief gap by S1, in Stage 1 and then to a backward protocol where the US was followed after a slightly shorter gap by S1 in Stage 2. They found that S2 subsequently elicited CRs, indicating that subjects had used the implied (short forward) relation between S2 and the US in Stage 2 to expect the US when tested with S2. This finding suggests a common ability to encode and use specific temporal relationships between events.

Temporal contiguity enhances learning in skin conductance (Morrow & Keough, 1968) and eyeblink conditioning protocols (e.g., Finkbiner & Woodruff-Pak, 1991), as well as in a paired associate task (Murray, 1970). As in animals, this effect is partly mediated by the reduced opportunity for interference from competing predictors at short CS–US intervals. Costa and Boakes (2011) included a number of distractor stimuli in an avoidance task that varied the duration of the interval between the warning signal and the negative outcome. They observed worse learning at longer intervals (weaker contiguity). A multiple regression analysis showed that the effect of interval duration on learning was primarily mediated by the number of distractors during that interval. Furthermore, learning (as measured by successful avoidance) was closely associated with the ability to identify *which* warning signal predicted the negative outcome. This finding is consistent with the evidence reviewed earlier regarding the necessity of contingency awareness for learning. Thus, a simple account of Costa and Boakes' (2011) findings is that distractor stimuli reduced the ability of participants to learn the warning signal—negative outcome contingency at longer intervals, which in turn led to poorer performance.

The effect of contiguity also appears to depend on whether learning is intentional (participants are instructed to look for relationships) or incidental. For example, in a paired associate learning task, Murray and Ure (1974) found better overall learning under intentional than incidental conditions. They also found the typical effect of contiguity (better learning with shorter interstimulus intervals) in the incidental condition but the opposite trend in the intentional condition. They attributed the latter finding to an increased opportunity for mediating activities such as rehearsal at longer intervals. Indeed, Weidemann et al. (2016) showed that incidental learning can fail to occur at all, even with high contiguity. They trained participants in two separate computer-based tasks and then asked them to perform the tasks on alternating trials with no intertrial interval. They embedded a stimulus-air puff contingency across the two tasks, such that the stimulus (CS) sometimes occurred at the end of one task, and when it did so, the air puff (US) reliably occurred at the start of the next task. Maintained performance to the primary tasks demonstrated that participants were attending to both the CS and the US. Under incidental conditions, however, eyeblink responding demonstrated that very few participants learned the CS-US association, despite attention to both stimuli, close contiguity between the stimuli, and a 100% reinforcement schedule. By contrast, participants given a hint that cross-task relations might exist showed much stronger learning on the eyeblink measure. Finally, across all participants, reliable eyeblink conditioning was only observed in participants who could verbally describe the contingency.

Similarly, when attention is drawn to the possibility of a relation between two events, participants can learn associations over long intervals (e.g., Greville et al., 2013). Furthermore, participants' expectations regarding the properties of a causal relationship can modulate the effects of contiguity. For example, Buehner and May (2004) found that judgments of causal strength depended on how closely the observed temporal relations between a cue and an outcome matched the causal model held by participants; the normal effect of contiguity could even be reversed when a stimulus-outcome interval was implausibly short for the causal scenario. This result is consistent with other evidence that learning is strongly influenced by prior beliefs about the specific relationship in question or by more abstract knowledge concerning the relationships between classes of stimuli or the mechanisms by which they exert their effects (Kemp & Tenenbaum, 2009; Lagnado et al., 2007).

Overall, truly incidental learning is difficult to detect, particularly with low contiguity or with competing/interfering stimuli. Conversely, such learning is more readily detected, even when contiguity is low, if participants' attention is drawn to the possibility of a relation by instructions, the structure of the task, or by exploiting prior causal knowledge of the stimuli involved. There is a parallel here to long delay learning in animals. Whereas animals require short CS–US intervals to show learning in most conditioning paradigms, they can learn cue-illness associations across intervals of several hours, providing the cue is a flavor. This "cue to consequence" effect has been explained in terms of an innate preparedness to associate gustatory cues with gustatory consequences (Seligman, 1970), which, in turn, can be viewed as an innate causal model. In terms of Revusky's (1971) interference model, focusing attention exclusively on flavors greatly reduces the number of cues that could compete for association with illness.

Expectancy Violation Drives Learning

Acquisition of Pavlovian conditioning in humans, as in animals, is governed by prediction error. For example, humans show phenomena that depend on the presence or absence of expectancy violation, such as blocking (Hinchy et al., 1995; Shanks, 1985), protection from extinction (Lovibond et al., 2000), and conditioned inhibition (Karazinov & Boakes, 2004; Neumann et al., 1997). Furthermore, these phenomena are strengthened by manipulations that make prediction error explicit. For example, when binary outcomes are used as USs (e.g., the presence or absence of shock or a symbolic outcome such as an allergic reaction), the status of the added cue (B) in a blocking design (A+ trials followed by AB+ trials) is ambiguous. Participants are unable to determine whether B does or does not predict the outcome because the magnitude of the outcome on AB trials cannot be any larger than that produced by A alone. This implies that allowing the outcome to be larger will remove the ambiguity and reveal blocking; if the outcome is the same for AB as it was for A, then B can be inferred to be ineffective (a form of counterfactual reasoning). This prediction has been confirmed in both causal learning and fear conditioning protocols using submaximal outcomes (De Houwer et al., 2002), additivity pretraining (Lovibond et al., 2003), and additivity instructions (Mitchell & Lovibond, 2002).

The US Is Encoded During Learning

The fact that humans can verbally report what they have learned about CS–US relations provides prima facie evidence that they have encoded the US. They also pass tests of US encoding used in animal research, for example, sensory preconditioning (White & Davey, 1989), blocking (De Houwer et al., 2002), US devaluation (Davey, 1989), and expectancy violation responses to surprising omission of a US (see Performance section). Together, these sources of evidence make it clear that people encode CS–US relationships and that this encoding underpins both verbal report and CRs.

Learning by Instruction Can Be Integrated With Learning by Experience

As noted earlier, verbal information can be used to convey the presence or absence of an association. These effects occur not only on self-report but also on behavioral and physiological measures of learning (Mertens et al., 2018). Importantly, common mechanisms underlie learning by instruction and experience. Lovibond (2003) used an "unovershadowing" protocol in which AB → shock trials were followed by A alone trials to assess whether a reduction in the associative strength of A retrospectively increased that of B (see Van Hamme & Wasserman, 1994). The critical finding was that upward revaluation of B was observed on both skin conductance and shock expectancy ratings, not only when the AB → shock and the A alone trials were both directly experienced (conditioning), or when both were instructed, but also when one was experienced and the other was instructed (Lovibond, 2003). A similar integration of instruction and direct experience has been reported in blocking, although without a physiological measure (Boddez, Baeyens, Hermans, et al., 2013). These results show that information acquired symbolically (by instruction) and through experience (by conditioning) is encoded in a way that allows them to be integrated, leading to greater self-reported shock expectancy and CRs to B. Because the unovershadowing effect relies on the prediction error arising from A alone trials, the increased responding to B suggests that the comparison between what is expected and what occurs takes place at an abstract representational level. These expectations are available for self-report, and they align with the strength of CRs.

Performance

The CR Is Not the Same as the UR

Humans, like animals, show CRs that are different from URs in some tasks and similar in others, making them hard to explain in terms of a US activation model. For example, a CS that signals a mildly painful shock elicits fear rather than pain, as indicated by elevations in startle responses, skin conductance, and self-reported fear/anxiety (Bach & Melinscak, 2020). However, in line with the animal eyeblink conditioning literature, as well as Konorski's (1967) preparatory/consummatory distinction, the CRs in human eyeblink conditioning closely parallel the UR (Woodruff-Pak & Steinmetz, 2000).

Expectancy Violation Impacts Behavior

Expectancy violation generates measurable changes in behavior in humans. For example, the "third interval response" in electrodermal conditioning occurs on nonreinforced test trials during the time period when the US would normally occur (e.g., Dunsmoor & LaBar, 2012; Prokasy & Ebel, 1967; Spoormaker et al., 2012). This is another example of a physiological response to the *nonoccurrence* of an expected event.

Expectancy Has Unique Stimulus Properties

In humans, as in animals, differential outcomes enhance the rate at which discrimination learning occurs (e.g., Mok & Overmier, 2007). We are not aware of any studies that have directly tested whether expectancies of a particular event can be discriminated from recent experience of that event, but it is reasonable to assume that humans can easily make that discrimination. Of course, in order to generate an expectancy of the US it predicts, a Pavlovian CS must be able to retrieve some sort of representation of the US from memory. However, that activation alone does not equate to causal or predictive strength; a person may believe the CS and US are associated without believing the CS causes or predicts the US. For example, in human causal learning, inhibitory stimuli activate a representation of the outcome whose absence they predict, as shown by speed of categorization, but they have the opposite effect on causal ratings (Mitchell et al., 2007). Similarly, a blocked cue is able to activate the outcome it has been paired with, as measured by cued recall, but it has little causal strength (Mitchell et al., 2005). Thus, retrieval of a US memory is necessary but not sufficient for human predictive or causal learning.

US Expectancy Ratings Track CRs

One of the most direct and compelling sources of evidence for a central role of expectancy in conditioning performance is the finding that self-reported US expectancy closely tracks learning and performance as indexed by other measures. For example, the development of discriminated US expectancies in a differential conditioning procedure is strongly related to the development of accurate explicit knowledge of the conditioning relationships, so-called contingency awareness (Lovibond et al., 2011; Purkis & Lipp, 2001; Schell et al., 1991). This relationship is not surprising, given that both are self-report measures. However, it contradicts the idea that expectancy ratings are based on the output of a separate implicit system, which predicts that expectancies would often arise in the absence of contingency knowledge (see the following section).

Furthermore, the development of CRs is closely related to the development of the corresponding expectancies. Dawson and Biferno (1973) used an ingenious design in which a differential contingency was embedded in an auditory discrimination task. Participants were told that their primary task was to discriminate between different tones under the stress of occasional electric shocks. They were also asked to rate their expectancy of shock so that this factor could be controlled for in statistical analyses of the data. This cover story slowed learning, as measured both by the development of electrodermal CRs and contingency awareness. Critically, for those participants who did become aware of the contingency, the onset of differential expectancies to the CS+ and CS− coincided with the onset of differential electrodermal responses to these stimuli. There was no evidence of differential responding on the electrodermal measure prior to the point of expectancy differentiation. This pattern of results has been replicated in the other primary human conditioning paradigm, eyeblink conditioning, with both single cue and differential procedures (Weidemann & Antees, 2012; Weidemann et al., 2013).

Expectancy ratings are highly reliable and sensitive, indicating that humans find it a natural and easily accessible construct to report. Boddez, Baeyens, Luyten, et al. (2013) reviewed the psychometric properties of US expectancy ratings in the context of fear conditioning, concluding that there is evidence for “the face validity, diagnostic validity, predictive validity and construct validity of the US-expectancy measure” (p. 201). Inconsistencies between expectancy ratings and behavioral or physiological CRs are generally quantitative rather than qualitative and can be accounted for either by the greater sensitivity of expectancy ratings or by method variance, that is, factors that are unique to a particular CR. For example, the skin conductance measure recorded during the CS in many fear conditioning experiments is subject to substantial habituation over trials, even when the CS is paired with an aversive US such as shock. Thus, the acquisition curve for self-reported expectancy typically shows a negatively accelerated upward trend for a CS paired with shock (CS+), while the curve for skin conductance typically increases for one or two trials and then declines. This factor is the main reason why a differential design with a nonshocked control stimulus (CS−) is commonly used in fear conditioning studies. Nonetheless, by the end of acquisition trials, the difference in skin conductance responding between CS+ and CS− is often very small, meaning that the skin conductance measure has little sensitivity at this point. This factor, along with the large individual differences between participants and high variability from trial to trial, also means that dissociations in which an effect is observed on expectancy but not skin conductance are not uncommon. Similar arguments apply to other physiological and behavioral CRs, and floor and ceiling effects on these measures also contribute. With due allowance for these factors, there is a strong relationship between self-reported expectancy and objective anticipatory CRs, both between individuals and within an individual over time (Lovibond & Shanks, 2002).

This strong relationship is consistent with a model in which expectancy plays a causal role in generating CRs. However, an alternative account of this relationship (e.g., Öhman & Mineka, 2001) denies such a causal role, instead proposing that the convergence between expectancy and anticipatory CRs derives from a retrospective attributional process whereby the participant infers US expectancy from their own anticipatory behavior. However, humans have little difficulty in rating their outcome expectancy in predictive and causal learning tasks, where the outcome is symbolic and induces no anticipatory behavior. They show exactly the same acquisition and extinction curves as are seen for expectancy ratings in fear conditioning tasks of the type which Öhman and Mineka used to base their proposal on (e.g., Humphreys, 1939; Ogallar et al., 2021). Therefore, we propose that a more parsimonious account of the close correspondence between self-reported expectancy and CRs is that verbalizable expectancy derives from introspective access to a distinct state of expectancy that is centrally involved in learning and performance.

Verbal Instruction Is Sufficient to Elicit Anticipatory Behavior

As noted earlier, verbal instruction can be used to convey an associative relation, and associations conveyed verbally can be combined with information acquired by experience (in conditioning protocols), strongly suggesting that instruction and experience generate associative knowledge in a similar form. However, it is also possible to *directly* generate an expectancy of a significant outcome without an intervening association between a stimulus and that outcome, for example, by means of a verbal threat manipulation with an aversive outcome such as electric shock. Here, informing a participant that they are about to receive a shock generates an immediate fear reaction as evidenced by self-report or physiological responses such as skin conductance, heart rate, or startle modulation (e.g., Buck et al., 1970; Grillon & Davis, 1997; Roxburgh et al., 2019). This finding is consistent with the idea that fear is an innate response to the expectancy of an upcoming aversive event, and critically, it does not matter if the expectancy was generated directly by instruction about an imminent event or via a mediating stimulus that had previously been associated with that event.

Challenges to the Role of Expectancy

We consider three phenomena commonly taken to challenge the idea that expectancy plays a causal role in Pavlovian conditioning, the first two empirical and the third theoretical.

The Perruchet Effect

As noted earlier, there is usually a close correspondence between Pavlovian CRs and US expectancy in humans. However, there is one prominent empirical finding that contradicts this pattern. Perruchet (1985) conducted an eyeblink study with a single tone CS that was followed by an air puff US on a random 50% of trials. Thus, participants were exposed to a series of trials that were either paired (tone-airpuff) or unpaired (tone alone), in random order. Before each tone trial they were asked to rate the degree to which they expected the airpuff to occur, and during the tone their eyeblink CRs were also recorded. Across the experiment, participants showed an intermediate level of eyeblink CRs to the tones, consistent with the 50% partial reinforcement schedule. However, when the data were sorted according to the length of runs of unpaired and paired trials, Perruchet observed a dissociation between eyeblink CRs and US expectancy. CRs were the strongest after a run of paired trials (e.g., four tone-airpuff trials) and the weakest after a run of unpaired trials (e.g., four tone alone trials), in accordance with recency of CS–US pairings. However, expectancy ratings showed the opposite pattern—participants expected the US most after a run of unpaired trials and least after a run of paired trials, in accordance with the gambler’s fallacy. This dissociation has also been observed with a reaction time priming version of the task (Perruchet et al., 2006).

At face value, this dissociation challenges an expectancy account of CR performance, which holds that US expectancy and CRs will be congruent. However, there is a potential bias in the Perruchet design in that recency of CS–US pairings is confounded with recency of US presentation. For example, a run of paired trials necessarily means the US has been experienced frequently in the recent past. Therefore, the stronger CRs observed after a run of paired trials could be due to nonassociative factors such as priming of the US representation or the response system. By unconfounding CS–US pairings and US recency, we have provided evidence that US recency can account for both the reaction time trend in the priming version of the task (Mitchell et al., 2010) and the CR trend in the eyeblink version (Weidemann & Lovibond, 2016). In other words, the trend in the CR measure over runs of trials does not appear to be due to a change in learning about the CS–US association but rather to the recency of experience of the US. This finding reduces the apparent double dissociation between CRs and expectancy to a *single* dissociation. We speculate that the expectancy trend could be due to an implicit demand effect due to participants being asked to give a rating on every trial, where the only factor that varies from trial to trial is the pattern of preceding trials. Although we believe the single dissociation between expectancy ratings and CRs in the Perruchet task should be explored further, we do not consider that it provides strong evidence against the idea that US expectancy drives CR performance.

The Role of the CS in Determining the Form of the CR

Although many different stimuli can serve as the CS in a variety of Pavlovian tasks, the nature of the CS plays a role in determining the observed CRs. For example, Holland (1977) reported that a visual CS for food elicits rearing in rats, whereas an auditory signal for food elicits a head-jerk response. The difficulty here is that if CRs are elicited solely by an expectancy of the US, then the type of CS that conveys the expectancy should be irrelevant; it is the message that should matter, not the messenger. The first point we would make is that this finding also poses a difficulty for a stimulus substitution model or any model that relies on activation of the US representation to produce CRs. Some associative models, for example, HEIDI (Honey et al., 2020), allow a backward US-CS association to elicit CS-related responses. However, the specific US employed still determines the CR observed, so both CS and US information must be integrated in some way. Within an expectancy model, certain CSs may already have an innate tendency to elicit (or release) a particular response, one that is augmented when that CS elicits an expectancy of a US from a matching motivational system. For example, Timberlake and Grant (1975) arranged that the introduction of a rat into a conditioning chamber signaled the delivery of food to a resident rat. They reported that the resident rat displayed prosocial behavior such as grooming toward the introduced rat rather than food-related behavior. Conversely, when the introduction of a lever signaled food delivery, the resident rat displayed food-related behaviors such as handling and biting the lever. A similar account could be offered for taste aversion learning, where an initially neutral taste paired with illness comes to elicit a distaste/rejection response similar to that shown to bitter tastes rather than the UR of nausea (Berridge, 2000).

Other features of a CS, such as its modality, location, and other affordances, also play a role in determining the CR. In pigeons, a localized visual CS that predicts delivery of food elicits approach and pecking responses directed toward the CS, a phenomenon known as sign tracking (Hearst & Jenkins, 1974). In contrast, a diffuse visual or auditory CS that predicts food elicits other behavior such as an increase in general activity, presumably because these stimuli are not sufficiently localized to support approach. Furthermore, in rats, when a localizable CS such as lever insertion is employed, some rats approach and contact the lever (sign tracking), while others approach the food magazine, a phenomenon known as goal tracking (Boakes, 1977). Sign tracking appears to be more likely with longer CS–US intervals and with partial reinforcement (Robinson et al., 2014; Timberlake et al., 1982), suggesting that it is a preparatory response, whereas goal tracking is a consummatory response (Konorski, 1967). Alternatively, goal tracking might involve Pavlovian enhancement of instrumental learning to approach the goal, a possibility made more plausible by the routine use of magazine approach training prior to Pavlovian conditioning. More importantly, for present purposes, Derman et al. (2018) have used US devaluation and cue competition to demonstrate that sign tracking and goal tracking share an underlying expectancy-based mechanism.

In summary, it seems that there are multiple environmental, temporal, and genetic factors that combine to determine which CRs will be triggered by the expectancy of a particular US. A systematic approach to understanding and predicting the form taken by the CR is the “behavior systems” model advocated by Timberlake (1984, 1994), in which field data provide the biological context within which conditioning and its expression in behavior can be understood. Although this approach requires analysis at the level of individual species and their environmental niche, it can be argued that this is the relevant level at which selection processes have operated and, hence, the relevant level to draw general conclusions about determinants of the form of CRs.

The Involuntary Nature of Pavlovian CRs

The final challenge to the idea that expectancy plays a causal role in Pavlovian conditioning is the involuntary nature of the CR, most dramatically illustrated in the so-called omission effect. This effect has been observed in salivary conditioning in dogs (Sheffield, 1965), jaw movement conditioning in rabbits (Gormezano & Hiller, 1972), and sign tracking in pigeons (Williams & Williams, 1969). In each of these protocols, the CS signaled a US *unless* the subject exhibited a CR, in which case the US was omitted. This instrumental contingency typically failed to eliminate the CR. For example, pigeons cycled through periods of pecking (thereby not receiving food) and ceasing to peck (thereby receiving food), suggesting that they were unable to inhibit their CRs in order to obtain food. Similar omission effects have been observed in humans. For example, Le Pelley et al. (2015) told participants they could earn money by quickly identifying a target stimulus in an array of distractors by looking directly at the target. One distractor was consistently related to a high reward for identifying the target. However, if participants looked at the distractor before the target, the reward was omitted. Despite this disincentive, participants found it difficult to refrain from looking at the high-value distractor. Le Pelley et al. (2015) interpreted their results as showing that the Pavlovian association between the distractor and reward elicited an attentional response to the distractor, which was involuntary in nature.

It is clear that the involuntary nature of CRs is correctly predicted by a reflexive model. However, it does not follow that the reflexive model is the *only* way in which the performance of Pavlovian CRs can be explained. As we have reviewed above, there is a great deal of empirical evidence that cannot easily be explained by a reflexive model, in particular qualitative differences between CRs and URs. At the same time, there is good evidence that expectancy states are elicited by CSs and play a critical role in CR performance. In humans, Pavlovian CRs closely track self-reported US expectancy, and, critically, the same behaviors that are elicited by CSs can be produced by verbal instruction either about a CS–US contingency or by direct instruction about the imminent occurrence of the US. The latter finding directly demonstrates that symbolic knowledge can elicit involuntary anticipatory behaviors. While we do not know *how* this occurs, we do know that it *does* occur. It therefore seems perverse to deny that, when such behaviors are elicited by a CS that has been directly paired with a US, they cannot be elicited through this same route.

The idea that a belief can activate emotional and behavioral response systems is uncontroversial in many other fields of psychology. For example, it is at the heart of cognitive appraisal theories of emotion (Lazarus & Folkman, 1984; Scherer & Moors, 2019). In particular, dominant theories of anxiety assume that anxiety is elicited by an expectancy of a future negative outcome, often decomposed into the perceived probability and cost of that outcome (e.g., Foa & Kozak, 1986; S. G. Hofmann, 2007). What we are proposing here is the parsimonious view that exactly the same mechanism can account for anxiety elicited by a stimulus that has predicted negative outcomes in the past—in other words, via Pavlovian conditioning.

Are CRs in animals also produced by expectancies? We have no way of determining the mental states of other species and no way of communicating a contingency to them in a symbolic form. However, throughout this article, we have noted the extensive convergence in the evidence for an expectancy-based mechanism in animals and humans. Certain findings, such as the qualitative differences in behaviors elicited by CSs and USs, seem to demand a mediating performance mechanism, and we suggest that the most plausible candidate for this mechanism is an anticipatory or expectancy state. This state likely differs in complexity between species, including humans, but the principles of evolution stress continuity. Indeed, it is this continuity that in part motivates the study of animal models in order to understand the functional and neural mechanisms of human psychological processes. Accordingly, we think it likely that precursors to the human expectancy state exist in other species and that the study of such states will be informative with regard to understanding anticipatory learning in both humans and other animals. Therefore, rather than continuing down a reflexive pathway with known limitations, we suggest instead that we turn our attention to trying to understand this state of expectancy and identify its underlying mechanisms.

Integrated Expectancy Model

Based on the work reviewed above, we suggest that Pavlovian conditioning involves learning a CS–US contingency that exists in the environment. In humans, this learning is verbally accessible and takes the form of a belief or rule (e.g., that the CS causes or predicts the US). In this regard, our position complements existing propositional models of learning (e.g., De Houwer, 2009; Mitchell et al., 2009a) but goes further in identifying a critical role for expectancy not only in learning but also in the performance of CRs. After learning, when the CS is subsequently encountered, it triggers a state of expectancy of the outcome. In humans, this state is also verbally accessible and takes the form of a belief that the outcome will occur in the near future. This expectancy state elicits involuntary anticipatory behavior appropriate to the US, such as emotional states, motor behavior, and physiological responses (CRs). These responses are innate and specific to the motivational system and the species in question. For example, in aversive conditioning, the CRs that are observed are all relevant to how that particular organism deals with threat—referred to by Bolles (1970) as species-specific defensive responses. These behaviors are often accompanied by activation of distinct emotional systems, such as fear, which support and coordinate behavioral responses. CRs often differ from URs, and they can also differ as a function of the temporal information conveyed by the signal. The adaptive advantages of such an arrangement are obvious—the organism can deploy behaviors appropriate to the future occurrence of biologically significant events based on their imminence rather than treating the signals as though they *are* those events.

In the above analysis, we ascribe a causal role to expectancy in generating anticipatory behavior, and we also suggest the expectancy is consciously accessible. However, we are not saying that the conscious experience per se is causal. Rather, it is possible and indeed likely that the brain state underlying expectancy gives rise to both anticipatory behavior and conscious experience, with the latter being epiphenomenal. Critically, however, introspective self-report provides a valid indicator of the underlying expectancy state that *is* responsible for CR performance. Our position differs radically from the reflexive model that is implicit even in contemporary error-prediction models of associative learning. These models acknowledge the role of expectancy in regulating learning, but they typically retain a simple excitatory/inhibitory link model to explain performance. Even temporal difference models (e.g., Sutton & Barto, 1981, 2018), which successfully account for many temporal factors in Pavlovian conditioning, retain a stimulus substitution architecture in which the CR is mediated by activity in nodes that also generate the UR. By contrast, we argue that a separate expectancy state plays a causal role in the performance of Pavlovian CRs by triggering anticipatory CRs that may differ qualitatively from the UR.

The position we advocate retains many of the strongest features of contemporary associative models, including the emphasis on contingency as well as contiguity, the role of prediction error in learning, and the involuntary nature of CR performance. However, it goes further in postulating a distinct expectancy state that underlies production of CRs. The field has already transitioned from S–R theories, which emphasized direct learning of responses, to the idea that we learn about S–S associations that exist in our environment (e.g., Mackintosh, 1983, Chapter 1). Our position is an extension of this theoretical transition to encompass the idea that CS–US associations are coded in an abstract rather than a reflexive way. This allows them to incorporate additional information, such as temporal parameters, and thereby elicit a wider range of innate anticipatory behaviors. It also means that the CS–US relationship can be accessed and used in other operations, such as reasoning, without automatically generating anticipatory behaviors. This in turn allows, at least in humans, for associative knowledge to be combined with other abstract knowledge, such as that transmitted verbally, in order to modulate expectancies and hence behavior (Lovibond, 2003; see Colloca & Miller, 2011, for a similar account of conditioning and instruction in placebo effects).

Finally, although the present article is concerned with Pavlovian conditioning, a parallel analysis can be put forward for instrumental conditioning. In this case, the relevant environmental contingency is between an action (a voluntary behavior) and an outcome. There is a great deal of evidence to suggest that a common learning mechanism underpins both Pavlovian and instrumental conditioning (e.g., Dickinson, 1980). However, the performance mechanisms are very different. Whereas Pavlovian CRs are elicited automatically when a CS activates expectancy of a US, deployment of instrumental knowledge involves a choice on the part of the organism. Here, the organism uses contingency knowledge to select an action with the strongest likelihood of gaining a desired outcome. This position offers a coherent account of the similarity in learning mechanism but the dissimilarity in performance mechanism between Pavlovian and instrumental conditioning.

There are two important ways in which Pavlovian and instrumental processes intersect. First, carrying out an action will itself activate a state of expectancy of the associated outcome and hence would be expected to produce the same anticipatory CRs as those elicited by Pavlovian CSs. Indeed, such CRs are often observed after an instrumental response has been performed (e.g., Williams, 1965). Furthermore, these anticipatory responses may occur as soon as the decision to perform an action has been made. Ng and Lovibond (2017) found exactly this pattern in human avoidance learning. Participants showed lower skin conductance response (indicative of reduced fear) in the period after they had indicated their intention to make an avoidance response but before they had actually performed the response. A second way in which Pavlovian and instrumental conditioning interact is in the Pavlovian instrumental transfer design. Here, a Pavlovian CS associated with a particular outcome augments instrumental responding based on the same outcome (e.g., Kruse et al., 1983). Activation of an outcome expectancy by the CS is a plausible mediator of this interaction (Hogarth, 2020).

Implications

Clinical

One of the most successful clinical applications of Pavlovian conditioning has been the development of exposure therapy for anxiety-eliciting stimuli (Craske et al., 2018). This application is based on the reasonable assumption that many anxiety reactions are due to a prior conditioning episode involving an association between a stimulus and some negative outcome. Within an expectancy framework, the stimulus elicits anxiety via the mediating mechanism of expectancy of the negative outcome (Lovibond, 2011). Exposure therapy is effective because the absence of any adverse outcome contradicts the patient's expectancy of harm and thereby leads to extinction—it weakens their belief concerning the connection between the stimulus and the negative outcome (Pittig et al., 2023). If patients fail to make this inference, exposure fails to confer any long-term benefit. For example, many individuals with a fear of flying suffer through repeated flights without symptom improvement (Nousi et al., 2008). This is why clinicians take steps to eliminate alternative explanations for the absence of harm, such as safety behaviors (e.g., Salkovskis et al., 1999), and why they are at pains to emphasize to patients the contradiction between their expectation and reality through behavioral experiments designed to test and falsify their excessive threat beliefs (Moulds et al., 2022). According to a reflexive view of conditioning and extinction, none of these verbal maneuvers should have any impact at all on anxiety established via conditioning.

An expectancy framework can also accommodate anxiety reactions established through other etiological pathways, such as observational learning and verbal instruction (Rachman, 1977). For example, when a child sees her father show fear to a novel stimulus, it is reasonable for the child to infer that the stimulus has great potential to cause harm. Similarly, when a person observes harm occurring to another person, such as witnessing an attack, it is reasonable for the observer to infer that a similar fate could befall them in the same circumstances. Verbal instruction can directly convey an expectancy, for example, when a parent explains to their child the dangers of crossing a busy road. Many subtler verbal and symbolic sources, such as peers, movies, and social media, can also contribute to a person's understanding of social and physical threats. Even anxiety reactions with a genetic basis, as appears to be the case for specific phobias such as fear of heights or water, may be mediated by expectancy of harm (Menzies & Clarke, 1993). The critical point here is that exposure can contradict an expectancy of harm, regardless of the source of that expectancy. And appropriately targeted verbal instruction can support the benefits of exposure, regardless of the source of the expectancy.

By contrast, exposure therapy has not been as successful in appetitive disorders such as drug addiction (Conklin & Tiffany, 2002). Here, clinicians arrange for drug-associated cues, which are known to elicit craving and trigger drug usage, to be presented in the absence of the drug, with the goal of encouraging extinction. The analysis of human Pavlovian conditioning presented here offers a potential account of why drug cue exposure does not confer significant clinical benefits in practice.[1] Cue exposure for both anxiety and addiction often occurs under somewhat artificial conditions, such as in the clinician's office or under their direct supervision. The difference between the two is that in the case of anxiety, the lack of association between a cue and its expected harmful outcome remains true when the patient leaves the therapeutic environment—clinicians typically only treat anxiety disorders that are irrational or excessive in objective terms. By contrast, the lack of association between drug-related cues and their anticipated drug effect does not hold outside the clinic—bars continue to offer alcohol, and cigarette packets continue to signal the availability of nicotine. Another approach is needed. An expectancy analysis based on causal beliefs offers a framework for the exploration of potential treatment approaches for both appetitive and aversive disorders with an associative basis.

The synergistic benefits of experience and language in exposure-based treatments for aversive disorders are also seen in clinical interventions based on instrumental learning. For example, parent training interventions encourage parents to use verbal labeling to clarify the reasons for behavior-contingent consequences such as reward and time-out (e.g., McMahon & Forehand, 2003). If the verbal component fails to align with reality—for example, when parents do not follow through on promised rewards or threatened penalties—the intervention is weakened. Historically, the progression from behavior therapy to cognitive behavior therapy over the past 50 years has led to a wide range of therapeutic interventions with both behavioral and verbal components. Unfortunately, these

[1] We are grateful to Oren Griffiths for suggesting this explanation.

components are often based on quite disparate theoretical rationales. Development of an integrative theoretical framework for cognitive-behavioral interventions has been impeded by a continued interpretation of behavioral components, particularly those based on Pavlovian conditioning such as exposure, as low level and divorced from cognition. A reconsideration of Pavlovian conditioning as expectancy-based may help us understand the ways in which Pavlovian principles can inform interventions and allow them to be combined with verbal and other therapeutic components in an optimal way. This recommendation will not come as any surprise to evidence-based clinicians, who have been combining verbal and behavioral interventions for decades. However, we include it here to counteract the dominant narrative in experimental psychology that conditioning-based techniques operate through a noncognitive, reflexive process and to emphasize the communality between an expectancy analysis and contemporary cognitive-behavioral practice.

Neuroscience

A great deal of research has been carried out in animals and humans to investigate the neural mechanisms that underlie Pavlovian conditioning. Since the advent of error correction models of conditioning with Rescorla and Wagner's (1972) model, progress has been made in identifying the neural correlates of prediction error. In particular, the activity of dopamine neurons in the striatum appears to correspond to both positive and negative prediction errors and even to the temporal pattern of prediction errors of the temporal difference model (Montague et al., 1996; Schultz & Romo, 1990; Sutton & Barto, 2018). However, much of this work has been done with instrumental procedures, and it is less clear what neural processes are involved in Pavlovian expectancy states in the anticipatory period prior to the US. Activity is seen in many brain regions during this period, including those related to sensory processing of the CS and US and production of CRs, as well as in other regions such as the frontal cortex, hippocampus, striatum, and midbrain (e.g., Pauli et al., 2012, 2019; Vitay & Hamker, 2014). Electroencephalographic recording in humans is also sensitive to expectancy violation, as seen, for example, in the phenomenon of mismatch negativity (Näätänen et al., 1978). Negative cortical waves such as the contingent negative variation and stimulus-preceding negativity have been suggested as markers of a general anticipatory state (Brunia et al., 2012).

There are several implications of an expectancy model of Pavlovian conditioning for the search for neural mechanisms. First, learning involves a representation of the relationship between a CS and US that encodes US identity and temporal information. This knowledge can be used directly for prediction but can also be combined with other information, as in sensory preconditioning and revaluation designs. These features suggest that learning about an association involves more than formation of an excitatory link between brain regions activated by the CS and US. For example, learning likely requires both the CS and US to be represented in working memory, thus involving other brain regions such as the frontal cortex (Aquino et al., 2023). The resulting associative knowledge may then be encoded in a distributed network such that it is available to other brain systems to be used flexibly to guide behavior (Gallistel & Balsam, 2014).

A second implication is that CR performance is driven by a specific US expectancy state that differs from the current experience of the US or a memory trace of the US. This state may activate a range of species-specific CRs that serve to prepare the organism for the US in question. Accordingly, one way to distinguish neural activity associated with US expectancy from activity associated with the US or UR may be to work backward from brain regions responsible for CRs that are distinct from URs. Another important question is whether there is a single central expectancy state or distinct states depending on the nature of the US. If the former, that state might be identified by activity in a common brain region during CSs paired with USs of different valences, including neutral ones. If the latter, expectancy states may be identified by common patterns of activation across different brain regions specific to the US employed. Varying CS modality and CS–US interval may also help isolate expectancy processing from stimulus and response processing.

In humans, expectancy is represented in consciousness and can be manipulated by language, offering additional opportunities for investigation. For example, self-reported expectancy can be used as reference point for identifying correlated neural activity in imaging studies. Direct instruction about an imminent US can activate an expectancy state without concomitant CS processing. Furthermore, advances in the neural analysis of cognitive processes such as working memory and reasoning may be applied to understanding their involvement in Pavlovian learning, storage, and performance. Similarities and differences in neural processing between humans and animals can then be used to better understand the evolutionary development of learning and memory.

Conclusions

In this article, we have reviewed evidence that Pavlovian conditioning involves an expectancy process that is critical for both learning *and* performance. The conventional view that performance is driven solely by the degree of activation in a US representation cannot explain qualitative differences between CRs and URs, the role of CS–US interval, or the functional differences between expecting, experiencing, and remembering the US. Direct evidence for the role of expectancy in CR performance is provided by responses elicited by expectancy violation and, in humans, by the ability of verbal instruction to generate CRs. We believe that an expectancy analysis will not only benefit our understanding of this important adaptive process but also enhance integration with adjacent fields such as cognition and emotion.

References

Amsel, A. (1992). *Frustration theory: An analysis of dispositional learning and memory*. Cambridge University Press. https://doi.org/10.1017/CBO9780511665561

Andreatta, M., & Pauli, P. (2017). Learning mechanisms underlying threat absence and threat relief: Influences of trait anxiety. *Neurobiology of Learning and Memory*, *145*, 105–113. https://doi.org/10.1016/j.nlm.2017.09.005

Aquino, T. G., Courellis, H., Mamelak, A. M., Rutishauser, U., & O'Doherty, J. P. (2023). *Encoding of predictive associations in human prefrontal and medial temporal neurons during Pavlovian conditioning*. bioRxiv. https://doi.org/10.1101/2023.02.10.528055

Arcediano, F., Escobar, M., & Miller, R. R. (2005). Bidirectional associations in humans and rats. *Journal of Experimental Psychology: Animal Behavior Processes*, *31*(3), 301–318. https://doi.org/10.1037/0097-7403.31.3.301

Atkinson, R. C., & Shiffrin, R. M. (1968). Human memory: A proposed system and its control processes. In K. W. Spence & J. T. Spence (Eds.), *The psychology of learning and motivation* (pp. 89–195). Academic Press.

Bach, D. R., & Melinscak, F. (2020). Psychophysiological modelling and the measurement of fear conditioning. *Behaviour Research and Therapy*, *127*, Article 103576. https://doi.org/10.1016/j.brat.2020.103576

Berridge, K. C. (2000). Measuring hedonic impact in animals and infants: Microstructure of affective taste reactivity patterns. *Neuroscience and Biobehavioral Reviews*, *24*(2), 173–198. https://doi.org/10.1016/S0149-7634(99)00072-X

Blanchard, R. J., & Blanchard, D. C. (1969). Crouching as an index of fear. *Journal of Comparative and Physiological Psychology*, *67*(3), 370–375. https://doi.org/10.1037/h0026779

Boakes, R. A. (1977). Performance on learning to associate a stimulus with positive reinforcement. In H. Davis & H. M. B. Hurwitz (Eds.), *Operant–Pavlovian interactions* (pp. 67–97). Lawrence Erlbaum Associates.

Boakes, R. A. (1989). How one might find evidence for conditioning in adult humans. In T. Archer & L.-G. Nilsson (Eds.), *Aversion, avoidance and anxiety: Perspectives on learning and memory* (pp. 381–402). Erlbaum.

Boakes, R. A., & Costa, D. S. J. (2014). Temporal contiguity in associative learning: Interference and decay from an historical perspective. *Journal of Experimental Psychology: Animal Learning and Cognition*, *40*(4), 381–400. https://doi.org/10.1037/xan0000040

Boddez, Y., Baeyens, F., Hermans, D., Van der Oord, S., & Beckers, T. (2013). Increasing the selectivity of threat through post-training instructions: Identifying one stimulus as source of danger reduces the threat value of surrounding stimuli. *Journal of Experimental Psychopathology*, *4*(4), 315–324. https://doi.org/10.5127/jep.028512

Boddez, Y., Baeyens, F., Luyten, L., Vansteenwegen, D., Hermans, D., & Beckers, T. (2013). Rating data are underrated: Validity of US expectancy in human fear conditioning. *Journal of Behavior Therapy and Experimental Psychiatry*, *44*(2), 201–206. https://doi.org/10.1016/j.jbtep.2012.08.003

Bolles, R. C. (1970). Species-specific defense reactions and avoidance learning. *Psychological Review*, *77*(1), 32–48. https://doi.org/10.1037/h0028589

Bolles, R. C. (1972). Reinforcement, expectancy and learning. *Psychological Review*, *79*(5), 394–409. https://doi.org/10.1037/h0033120

Bolles, R. C., & Fanselow, M. S. (1980). A perceptual-defensive-recuperative model of fear and pain. *Behavioral and Brain Sciences*, *3*(2), 291–301. https://doi.org/10.1017/S0140525X0000491X

Brewer, W. F. (1974). There is no convincing evidence for operant or classical conditioning in adult humans. In W. B. Weimer & D. S. Palermo (Eds.), *Cognition and the symbolic processes* (pp. 1–42). Erlbaum.

Brunia, C. H. M., van Boxtel, G. J. M., & Böcker, K. B. E. (2012). Negative slow waves as indices of anticipation: The Bereitschaftspotential, the contingent negative variation, and the stimulus-preceding negativity. In S. J. Luck & E. S. Kappenman (Eds.), *The Oxford handbook of event-related potential components* (pp. 189–207). Oxford University Press.

Buck, R. W., Parke, R. D., & Buck, M. (1970). Skin conductance, heart rate, and attention to the environment in two stressful situations. *Psychonomic Science*, *18*(2), 95–96. https://doi.org/10.3758/BF03335714

Buehner, M. J., & May, J. (2004). Abolishing the effect of reinforcement delay on human causal learning. *The Quarterly Journal of Experimental Psychology B: Comparative and Physiological Psychology*, *57*(2b), 179–191. https://doi.org/10.1080/02724990344000123

Carter, R. M., Hofstötter, C., Tsuchiya, N., & Koch, C. (2003). Working memory and fear conditioning. *Proceedings of the National Academy of Sciences of the United States of America*, *100*(3), 1399–1404. https://doi.org/10.1073/pnas.0334049100

Clark, R. E., & Squire, L. R. (1998). Classical conditioning and brain systems: The role of awareness. *Science*, *280*(5360), 77–81. https://doi.org/10.1126/science.280.5360.77

Colloca, L., & Miller, F. G. (2011). How placebo responses are formed: A learning perspective. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *366*(1572), 1859–1869. https://doi.org/10.1098/rstb.2010.0398

Conklin, C. A., & Tiffany, S. T. (2002). Applying extinction research and theory to cue-exposure addiction treatments. *Addiction*, *97*(2), 155–167. https://doi.org/10.1046/j.1360-0443.2002.00014.x

Costa, D. S., & Boakes, R. A. (2011). Varying temporal contiguity and interference in a human avoidance task. *Journal of Experimental Psychology: Animal Behavior Processes*, *37*(1), 71–78. https://doi.org/10.1037/a0021192

Courville, A. C., Daw, N. D., & Touretzky, D. S. (2006). Bayesian theories of conditioning in a changing world. *Trends in Cognitive Sciences*, *10*(7), 294–300. https://doi.org/10.1016/j.tics.2006.05.004

Craske, M. G., Hermans, D., & Vervliet, B. (2018). State-of-the-art and future directions for extinction as a translational model for fear and anxiety. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *373*(1742), Article 20170025. https://doi.org/10.1098/rstb.2017.0025

Crespi, L. P. (1942). Quantitative variation of incentive and performance in the white rat. *The American Journal of Psychology*, *55*(4), 467–517. https://doi.org/10.2307/1417120

Davey, G. C. (1989). UCS revaluation and conditioning models of acquired fears. *Behaviour Research and Therapy*, *27*(5), 521–528. https://doi.org/10.1016/0005-7967(89)90086-7

Dawson, M. E., & Biferno, M. A. (1973). Concurrent measurement of awareness and electrodermal classical conditioning. *Journal of Experimental Psychology*, *101*(1), 55–62. https://doi.org/10.1037/h0035524

Dawson, M. E., & Schell, A. M. (1985). Information processing and human autonomic classical conditioning. In P. K. Ackles, J. R. Jennings, & M. G. H. Coles (Eds.), *Advances in psychophysiology* (Vol. 1, pp. 89–165). JAI Press.

Dayan, P., & Berridge, K. C. (2014). Model-based and model-free Pavlovian reward learning: Revaluation, revision, and revelation. *Cognitive, Affective & Behavioral Neuroscience*, *14*(2), 473–492. https://doi.org/10.3758/s13415-014-0277-8

De Houwer, J. (2009). The propositional approach to associative learning as an alternative for association formation models. *Learning & Behavior*, *37*(1), 1–20. https://doi.org/10.3758/LB.37.1.1

De Houwer, J., Beckers, T., & Glautier, S. (2002). Outcome and cue properties modulate blocking. *The Quarterly Journal of Experimental Psychology A: Human Experimental Psychology*, *55*(3), 965–985. https://doi.org/10.1080/02724980143000578

Derman, R. C., Schneider, K., Juarez, S., & Delamater, A. R. (2018). Sign-tracking is an expectancy-mediated behavior that relies on prediction error mechanisms. *Learning & Memory*, *25*(10), 550–563. https://doi.org/10.1101/lm.047365.118

DeVito, P. L., & Fowler, H. (1987). Enhancement of conditioned inhibition via an extinction treatment. *Animal Learning & Behavior*, *15*(4), 448–454. https://doi.org/10.3758/BF03205055

Dickinson, A. (1980). *Contemporary animal learning theory*. Cambridge University Press.

Dickinson, A. (1989). Expectancy theory in animal conditioning. In S. B. Klein & R. R. Mowrer (Eds.), *Contemporary learning theories: Pavlovian conditioning and the status of traditional learning theory* (pp. 279–308). Lawrence Erlbaum Associates.

Dickinson, A., & Shanks, D. (1985). Animal conditioning and human causality judgment. In L. Nilsson & T. Archer (Eds.), *Perspectives on learning and memory* (pp. 167–191). Lawrence Erlbaum Associates.

Dunsmoor, J. E., & LaBar, K. S. (2012). Brain activity associated with omission of an aversive event reveals the effects of fear learning and generalization. *Neurobiology of Learning and Memory*, *97*(3), 301–312. https://doi.org/10.1016/j.nlm.2012.02.003

Eikelboom, R., & Stewart, J. (1982). Conditioning of drug-induced physiological responses. *Psychological Review*, *89*(5), 507–528. https://doi.org/10.1037/0033-295X.89.5.507

Fanselow, M. S., & Lester, L. S. (1988). A functional behavioristic approach to aversively motivated behavior: Predatory imminence as a determinant of the topography of defensive behavior. In R. C. Bolles & M. D. Beecher (Eds.), *Evolution and learning* (pp. 185–212). Erlbaum.

Finkbiner, R. G., & Woodruff-Pak, D. S. (1991). Classical eyeblink conditioning in adulthood: Effects of age and interstimulus interval on acquisition in the trace paradigm. *Psychology and Aging*, *6*(1), 109–117. https://doi.org/10.1037/0882-7974.6.1.109

Flaherty, C. F. (1982). Incentive contrast: A review of behavioral changes following shifts in reward. *Animal Learning & Behavior*, *10*(4), 409–440. https://doi.org/10.3758/BF03212282

Foa, E. B., & Kozak, M. J. (1986). Emotional processing of fear: Exposure to corrective information. *Psychological Bulletin*, *99*(1), 20–35. https://doi.org/10.1037/0033-2909.99.1.20

Gallistel, C. R., & Balsam, P. D. (2014). Time to rethink the neural mechanisms of learning and memory. *Neurobiology of Learning and Memory*, *108*, 136–144. https://doi.org/10.1016/j.nlm.2013.11.019

Gawronski, B., & Mitchell, D. G. V. (2014). Simultaneous conditioning of valence and arousal. *Cognition and Emotion*, *28*(4), 577–595. https://doi.org/10.1080/02699931.2013.843506

Gewirtz, J. C., Brandon, S. E., & Wagner, A. R. (1998). Modulation of the acquisition of the rabbit eyeblink conditioned response by conditioned contextual stimuli. *Journal of Experimental Psychology: Animal Behavior Processes*, *24*(1), 106–117. https://doi.org/10.1037/0097-7403.24.1.106

Gormezano, I., & Hiller, G. W. (1972). Omission training of the jaw movement response of the rabbit to a water US. *Psychonomic Science*, *29*(5), 276–278. https://doi.org/10.3758/BF03329204

Gormezano, I., & Kehoe, E. J. (1981). Classical conditioning and the Law of Contiguity. In P. Harzem & M. D. Zelier (Eds.), *Predictability, correlation, and contiguity* (pp. 1–45). Wiley.

Gormezano, I., Schneiderman, N., Deaux, E., & Fuentes, I. (1962). Nictitating membrane: Classical conditioning and extinction in the albino rabbit. *Science*, *138*(3536), 33–34. https://doi.org/10.1126/science.138.3536.33

Green, L. J. S., Luck, C. C., Gawronski, B., & Lipp, O. V. (2021). Contrast effects in backward evaluative conditioning: Exploring effects of affective relief/disappointment versus instructional information. *Emotion*, *21*(2), 350–359. https://doi.org/10.1037/emo0000701

Greville, W. J., Cassar, A. A., Johansen, M. K., & Buehner, M. J. (2013). Structural awareness mitigates the effect of delay in human causal learning. *Memory & Cognition*, *41*(6), 904–916. https://doi.org/10.3758/s13421-013-0308-7

Grillon, C., & Davis, M. (1997). Effects of stress and shock anticipation on prepulse inhibition of the startle reflex. *Psychophysiology*, *34*(5), 511–517. https://doi.org/10.1111/j.1469-8986.1997.tb01737.x

Guthrie, E. R. (1935). *The psychology of learning*. Harper.

Harris, J. A., Westbrook, R. F., Duffield, T. Q., & Bentivoglio, M. (1995). Fos expression in the spinal cord is suppressed in rats displaying conditioned hypoalgesia. *Behavioral Neuroscience*, *109*(2), 320–328. https://doi.org/10.1037/0735-7044.109.2.320

Hearst, E., & Jenkins, H. (1974). *Sign tracking: The stimulus-reinforcer relation and directed action*. Psychonomic Society.

Hinchy, J., Lovibond, P. F., & Ter-Horst, K. M. (1995). Blocking in human electrodermal conditioning. *The Quarterly Journal of Experimental Psychology Section B*, *48*(1B), 2–12. https://journals.sagepub.com/doi/abs/10.1080/14640749508401433

Hofmann, S. G. (2007). Cognitive factors that maintain social anxiety disorder: A comprehensive model and its treatment implications. *Cognitive Behaviour Therapy*, *36*(4), 193–209. https://doi.org/10.1080/16506070701421313

Hofmann, W., De Houwer, J., Perugini, M., Baeyens, F., & Crombez, G. (2010). Evaluative conditioning in humans: A meta-analysis. *Psychological Bulletin*, *136*(3), 390–421. https://doi.org/10.1037/a0018916

Hogarth, L. (2020). Addiction is driven by excessive goal-directed drug choice under negative affect: Translational critique of habit and compulsion theory. *Neuropsychopharmacology*, *45*(5), 720–735. https://doi.org/10.1038/s41386-020-0600-8

Holland, P. C. (1977). Conditioned stimulus as a determinant of the form of the Pavlovian conditioned response. *Journal of Experimental Psychology: Animal Behavior Processes*, *3*(1), 77–104. https://doi.org/10.1037/0097-7403.3.1.77

Holland, P. C. (1980). CS-US interval as a determinant of the form of Pavlovian appetitive conditioned responses. *Journal of Experimental Psychology: Animal Behavior Processes*, *6*(2), 155–174. https://doi.org/10.1037/0097-7403.6.2.155

Holland, P. C., & Rescorla, R. A. (1975). The effect of two ways of devaluing the unconditioned stimulus after first- and second-order appetitive conditioning. *Journal of Experimental Psychology: Animal Behavior Processes*, *1*(4), 355–363. https://doi.org/10.1037/0097-7403.1.4.355

Holland, P. C., & Straub, J. J. (1979). Differential effects of two ways of devaluing the unconditioned stimulus after Pavlovian appetitive conditioning. *Journal of Experimental Psychology: Animal Behavior Processes*, *5*(1), 65–78. https://doi.org/10.1037/0097-7403.5.1.65

Holmes, N. M., Wong, F. S., Bouchekioua, Y., & Westbrook, R. F. (2022). Not "either-or" but "which-when": A review of the evidence for integration in sensory preconditioning. *Neuroscience and Biobehavioral Reviews*, *132*, 1197–1204. https://doi.org/10.1016/j.neubiorev.2021.10.032

Honey, R. C., Dwyer, D. M., & Iliescu, A. F. (2020). HeiDI: A model for Pavlovian learning and performance with reciprocal associations. *Psychological Review*, *127*(5), 829–852. https://doi.org/10.1037/rev0000196

Hull, C. (1943). *Principles of behavior: An introduction to behavior theory*. Appleton-Century-Crofts.

Humphreys, L. G. (1939). Acquisition and extinction of verbal expectations in a situation analogous to conditioning. *Journal of Experimental Psychology*, *25*(3), 294–301. https://doi.org/10.1037/h0053555

Jozefowiez, J. (2018). Associative versus predictive processes in Pavlovian conditioning. *Behavioural Processes*, *154*, 21–26. https://doi.org/10.1016/j.beproc.2017.12.016

Kamin, L. J. (1969). Predictability, surprise, attention and conditioning. In B. A. Campbell & R. M. Church (Eds.), *Punishment and aversive behavior* (pp. 279–296). Appleton-Century-Crofts.

Karazinov, D. M., & Boakes, R. A. (2004). Learning about cues that prevent an outcome: Conditioned inhibition and differential inhibition in human predictive learning. *The Quarterly Journal of Experimental Psychology B: Comparative and Physiological Psychology*, *57*(2b), 153–178. https://doi.org/10.1080/02724990344000033

Kemp, C., & Tenenbaum, J. B. (2009). Structured statistical models of inductive reasoning. *Psychological Review*, *116*(1), 20–58. https://doi.org/10.1037/a0014282

Kimble, G. A. (1947). Conditioning as a function of the time between conditioned and unconditioned stimuli. *Journal of Experimental Psychology*, *37*(1), 1–15. https://doi.org/10.1037/h0059371

Kirsch, I., Lynn, S. J., Vigorito, M., & Miller, R. R. (2004). The role of cognition in classical and operant conditioning. *Journal of Clinical Psychology*, *60*(4), 369–392. https://doi.org/10.1002/jclp.10251

Klosterhalfen, S., Rüttgers, A., Krumrey, E., Otto, B., Stockhorst, U., Riepl, R. L., Probst, T., & Enck, P. (2000). Pavlovian conditioning of taste aversion using a motion sickness paradigm. *Psychosomatic Medicine*, *62*(5), 671–677. https://doi.org/10.1097/00006842-200009000-00011

Konorski, J. (1967). *Integrative activity of the brain*. University of Chicago Press.

Kruse, J. M., Overmier, J. B., Konz, W. A., & Rokke, E. (1983). Pavlovian conditioned stimulus effects upon instrumental choice behavior are reinforcer specific. *Learning and Motivation*, *14*(2), 165–181. https://doi.org/10.1016/0023-9690(83)90004-8

Lagnado, D. A., Waldmann, M. R., Hagmaye, Y., & Sloman, S. A. (2007). Beyond covariation: Cues to causal structure. In A. Gopnik & L. E. Schulz (Eds.), *Causal learning: Psychology, philosophy, and computation* (pp. 154–172). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195176803.003.0011

Lazarus, R. S., & Folkman, S. (1984). *Stress, appraisal, and coping*. Springer.

Le Pelley, M. E., Pearson, D., Griffiths, O., & Beesley, T. (2015). When goals conflict with values: Counterproductive attentional and oculomotor capture by reward-related stimuli. *Journal of Experimental Psychology: General*, *144*(1), 158–171. https://doi.org/10.1037/xge0000037

Lee, J. C., Hayes, B. K., & Lovibond, P. F. (2018). Peak shift and rules in human generalization. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *44*(12), 1955–1970. https://doi.org/10.1037/xlm0000558

Lee, J. C., Lovibond, P. F., Hayes, B. K., & Navarro, D. J. (2019). Negative evidence and inductive reasoning in generalization of associative learning. *Journal of Experimental Psychology: General*, *148*(2), 289–303. https://doi.org/10.1037/xge0000496

Linwick, D. C., & Overmier, J. B. (2006). Associatively activated representations of food events resemble food outcome expectancies more closely than they resemble food-based memories. *Learning & Behavior*, *34*(1), 1–12. https://doi.org/10.3758/BF03192866

Lovibond, P. F. (1980). Effects of long- and variable-duration signals for food on activity, instrumental responding and eating. *Learning and Motivation*, *11*(2), 164–184. https://doi.org/10.1016/0023-9690(80)90011-9

Lovibond, P. F. (2003). Causal beliefs and conditioned responses: Retrospective revaluation induced by experience and by instruction. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *29*(1), 97–106. https://doi.org/10.1037/0278-7393.29.1.97

Lovibond, P. F. (2011). Learning and anxiety: A cognitive perspective. In T. R. Schachtman & S. S. Reilly (Eds.), *Associative learning and conditioning theory: Human and non-human applications* (pp. 104–120). Oxford University Press. https://doi.org/10.1093/acprof:oso/9780199735969.003.0037

Lovibond, P. F., Davis, N. R., & O'Flaherty, A. S. (2000). Protection from extinction in human fear conditioning. *Behaviour Research and Therapy*, *38*(10), 967–983. https://doi.org/10.1016/S0005-7967(99)00121-7

Lovibond, P. F., Liu, J. C., Weidemann, G., & Mitchell, C. J. (2011). Awareness is necessary for differential trace and delay eyeblink conditioning in humans. *Biological Psychology*, *87*(3), 393–400. https://doi.org/10.1016/j.biopsycho.2011.05.002

Lovibond, P. F., & Shanks, D. R. (2002). The role of awareness in Pavlovian conditioning: Empirical evidence and theoretical implications. *Journal of Experimental Psychology: Animal Behavior Processes*, *28*(1), 3–26. https://doi.org/10.1037/0097-7403.28.1.3

Lovibond, P. F., Been, S., Mitchell, C. J., Bouton, M. E., & Frohardt, R. (2003). Forward and backward blocking of causal judgment is enhanced by additivity of effect magnitude. *Memory and Cognition*, *31*, 133–142. https://doi.org/10.3758/BF03196088

Luck, C. C., & Lipp, O. V. (2017). Startle modulation and explicit valence evaluations dissociate during backward fear conditioning. *Psychophysiology*, *54*(5), 673–683. https://doi.org/10.1111/psyp.12834

Mackintosh, N. J. (1983). *Conditioning and associative learning*. Clarendon Press.

Málková, L., Wackermann, J., Hrudová, L., & Dostálek, C. (1989). Strictly simultaneous classical eyelid conditioning in man. *Activitas Nervosa Superior*, *31*(1), 7–15.

McEchron, M. D., McCabe, P. M., Green, E. J., Llabre, M. M., & Schneiderman, N. (1992). Air puff versus shock unconditioned stimuli in rabbit heart rate conditioning. *Physiology & Behavior*, *51*(1), 195–199. https://doi.org/10.1016/0031-9384(92)90223-O

McLaren, I. P. L., Forrest, C. L. D., McLaren, R. P., Jones, F. W., Aitken, M. R. F., & Mackintosh, N. J. (2014). Associations and propositions: The case for a dual-process account of learning in humans. *Neurobiology of Learning and Memory*, *108*, 185–195. https://doi.org/10.1016/j.nlm.2013.09.014

McMahon, R. J., & Forehand, R. L. (2003). *Helping the noncompliant child: Family-based treatment for oppositional behavior* (2nd ed.). Guilford Press.

McNally, G. P., & Westbrook, R. F. (1998). Effects of systemic, intracerebral, or intrathecal administration of an N-methyl-D-aspartate receptor antagonist on associative morphine analgesic tolerance and hyperalgesia in rats. *Behavioral Neuroscience*, *112*(4), 966–978. https://doi.org/10.1037/0735-7044.112.4.966

Menzies, R. G., & Clarke, J. C. (1993). The etiology of fear of heights and its relationship to severity and individual response patterns. *Behaviour Research and Therapy*, *31*(4), 355–365. https://doi.org/10.1016/0005-7967(93)90093-A

Mertens, G., Boddez, Y., Sevenster, D., Engelhard, I. M., & De Houwer, J. (2018). A review on the effects of verbal instructions in human fear conditioning: Empirical findings, theoretical considerations, and future directions. *Biological Psychology*, *137*, 49–64. https://doi.org/10.1016/j.biopsycho.2018.07.002

Mertens, G., & Engelhard, I. M. (2020). A systematic review and meta-analysis of the evidence for unaware fear conditioning. *Neuroscience and Biobehavioral Reviews*, *108*, 254–268. https://doi.org/10.1016/j.neubiorev.2019.11.012

Mitchell, C. J., De Houwer, J., & Lovibond, P. F. (2009a). The propositional nature of human associative learning. *Behavioral and Brain Sciences*, *32*(2), 183–198. https://doi.org/10.1017/S0140525X09000855

Mitchell, C. J., De Houwer, J., & Lovibond, P. F. (2009b). Link-based learning theory creates more problems than it solves. *Behavioral and Brain Sciences*, *32*(2), 230–246. https://doi.org/10.1017/S0140525X09001186

Mitchell, C. J., Livesey, E., & Lovibond, P. F. (2007). A dissociation between causal judgement and the ease with which a cause is categorized with its effect. *The Quarterly Journal of Experimental Psychology*, *60*(3), 400–417. https://doi.org/10.1080/17470210601002512

Mitchell, C. J., & Lovibond, P. F. (2002). Backward and forward blocking in human electrodermal conditioning: Blocking requires an assumption of outcome additivity. *The Quarterly Journal of Experimental Psychology B: Comparative and Physiological Psychology*, *55*(4b), 311–329. https://doi.org/10.1080/02724990244000025

Mitchell, C. J., Lovibond, P. F., & Gan, C. Y. (2005). A dissociation between causal judgment and outcome recall. *Psychonomic Bulletin & Review*, *12*(5), 950–954. https://doi.org/10.3758/BF03196791

Mitchell, C. J., Wardle, S. G., Lovibond, P. F., Weidemann, G., & Chang, B. P. I. (2010). Do reaction times in the Perruchet effect reflect variations in the strength of an associative link? *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *36*(2), 567–572. https://doi.org/10.1037/a0018433

Mok, L. W., & Overmier, J. B. (2007). The differential outcomes effect in normal human adults using a concurrent-task within-subjects design and sensory outcomes. *The Psychological Record*, *57*(2), 187–200. https://doi.org/10.1007/BF03395571

Montague, P. R., Dayan, P., & Sejnowski, T. J. (1996). A framework for mesencephalic dopamine systems based on predictive Hebbian learning. *The Journal of Neuroscience*, *16*(5), 1936–1947. https://doi.org/10.1523/JNEUROSCI.16-05-01936.1996

Morrow, M. C., & Keough, T. E., III. (1968). GSR conditioning with long interstimulus intervals. *Journal of Experimental Psychology*, *77*(3, Pt. 1), 460–467. https://doi.org/10.1037/h0025941

Moscovitch, A., & LoLordo, V. M. (1968). Role of safety in the Pavlovian backward fear conditioning procedure. *Journal of Comparative and Physiological Psychology*, *66*(3, Pt. 1), 673–678. https://doi.org/10.1037/h0026548

Moulds, M. L., Grisham, J. R., & Graham, B. M. (2022). *Cognitive behavioral therapy for anxiety*. Oxford Research Encyclopedias. https://doi.org/10.1093/acrefore/9780190236557.013.331

Murray, H. G. (1970). Incidental paired-associate learning as a function of interstimulus interval. *Journal of Verbal Learning and Verbal Behavior*, *9*(6), 642–646. https://doi.org/10.1016/S0022-5371(70)80026-3

Murray, H. G., & Ure, G. (1974). Role of temporal contiguity in intentional and incidental paired-associate learning. *Journal of Experimental Psychology*, *103*(4), 816–820. https://doi.org/10.1037/h0037177

Näätänen, R., Gaillard, A. W. K., & Mäntysalo, S. (1978). Early selective-attention effect on evoked potential reinterpreted. *Acta Psychologica*, *42*(4), 313–329. https://doi.org/10.1016/0001-6918(78)90006-9

Neumann, D. L., Lipp, O. V., & Siddle, D. A. T. (1997). Conditioned inhibition of autonomic Pavlovian conditioning in humans. *Biological Psychology*, *46*(3), 223–233. https://doi.org/10.1016/S0301-0511(97)05248-4

Ng, A., & Lovibond, P. F. (2017). Intentions matter: Avoidance intentions regulate anxiety via outcome expectancy. *Behaviour Research and Therapy*, *96*, 57–65. https://doi.org/10.1016/j.brat.2017.03.007

Nousi, A., Haringsma, R., Van Gerwen, L. J., & Spinhoven, P. (2008). Different flying histories in flying phobics: Association with psychopathology and treatment outcome. *Aviation, Space, and Environmental Medicine*, *79*(10), 953–959. https://doi.org/10.3357/ASEM.2189.2008

Ogallar, P. M., Callejas-Aguilera, J. E., Rosas, J. M., & Lamoureux, J. A. (2021). Concurrent evidence of extinction making acquisition context specific and ABA and ABC renewal effects in human predictive learning. *Journal of Experimental Psychology: Animal Learning and Cognition*, *47*(2), 137–149. https://doi.org/10.1037/xan0000288

Öhman, A., & Mineka, S. (2001). Fears, phobias, and preparedness: Toward an evolved module of fear and fear learning. *Psychological Review*, *108*(3), 483–522. https://doi.org/10.1037/0033-295X.108.3.483

Papini, M. R., & Dudley, R. T. (1997). Consequences of surprising reward omissions. *Review of General Psychology*, *1*(2), 175–197. https://doi.org/10.1037/1089-2680.1.2.175

Pauli, W. M., Gentile, G., Collette, S., Tyszka, J. M., & O'Doherty, J. P. (2019). Evidence for model-based encoding of Pavlovian contingencies in the human brain. *Nature Communications*, *10*(1), Article 1099. https://doi.org/10.1038/s41467-019-08922-7

Pauli, W. M., Hazy, T. E., & O'Reilly, R. C. (2012). Expectancy, ambiguity, and behavioral flexibility: Separable and complementary roles of the orbital frontal cortex and amygdala in processing reward expectancies. *Journal of Cognitive Neuroscience*, *24*(2), 351–366. https://doi.org/10.1162/jocn_a_00155

Pavlov, I. P. (1941). *Lectures on conditioned reflexes. Vol. 2. Conditioned reflexes and psychiatry*. International Publishers.

Pavlov, I. P. (1927). *Conditioned reflexes* (G. V. Anrep, Trans.). Oxford University Press.

Pearce, J. M., & Hall, G. (1980). A model for Pavlovian learning: Variations in the effectiveness of conditioned but not of unconditioned stimuli. *Psychological Review*, *87*(6), 532–552. https://doi.org/10.1037/0033-295X.87.6.532

Perruchet, P. (1985). A pitfall for the expectancy theory of human eyelid conditioning. *Pavlovian Journal of Biological Science*, *20*(4), 163–170. https://doi.org/10.1007/BF03003653

Perruchet, P., Cleeremans, A., & Destrebecqz, A. (2006). Dissociating the effects of automatic activation and explicit expectancy on reaction times in a simple associative learning task. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, *32*(5), 955–965. https://doi.org/10.1037/0278-7393.32.5.955

Peterson, G. B., Linwick, D., & Overmier, J. B. (1987). On the comparative efficacy of memories and expectancies as cues for choice behavior in pigeons. *Learning and Motivation*, *18*(1), 1–20. https://doi.org/10.1016/0023-9690(87)90020-8

Pittig, A., Heinig, I., Goerigk, S., Richter, J., Hollandt, M., Lueken, U., Pauli, P., Deckert, J., Kircher, T., Straube, B., Neudeck, P., Koelkebeck, K., Dannlowski, U., Arolt, V., Fydrich, T., Fehm, L., Ströhle, A., Totzeck, C., Margraf, J., … Wittchen, H.-U. (2023). Change of threat expectancy as mechanism of exposure-based psychotherapy for anxiety disorders: Evidence from 8,484 exposure exercises of 605 patients. *Clinical Psychological Science*, *11*(2), 199–217. https://doi.org/10.1177/21677026221101379

Powell, D. A., & Levine-Bryce, D. (1988). A comparison of two model systems of associative learning: Heart rate and eyeblink conditioning in the rabbit. *Psychophysiology*, *25*(6), 672–682. https://doi.org/10.1111/j.1469-8986.1988.tb01906.x

Prével, A., Rivière, V., Darcheville, J., & Urcelay, G. P. (2016). Conditioned reinforcement and backward association. *Learning and Motivation*, *56*, 38–47. https://doi.org/10.1016/j.lmot.2016.09.004

Prokasy, W. F., & Ebel, H. C. (1967). Three components of the classically conditioned GSR in human subjects. *Journal of Experimental Psychology*, *73*(2), 247–256. https://doi.org/10.1037/h0024108

Purkis, H. M., & Lipp, O. V. (2001). Does affective learning exist in the absence of contingency awareness? *Learning and Motivation*, *32*(1), 84–99. https://doi.org/10.1006/lmot.2000.1066

Rachman, S. (1977). The conditioning theory of fear-acquisition: A critical examination. *Behaviour Research and Therapy*, *15*(5), 375–387. https://doi.org/10.1016/0005-7967(77)90041-9

Ratner, S. C. (1967). Comparative aspects of hypnosis. In J. E. Gordon (Ed.), *Handbook of clinical and experimental hypnosis* (pp. 550–585). Macmillan.

Rescorla, R. A. (1969). Pavlovian conditioned inhibition. *Psychological Bulletin*, *72*(2), 77–94. https://doi.org/10.1037/h0027760

Rescorla, R. A. (1973). Effect of US habituation following conditioning. *Journal of Comparative and Physiological Psychology*, *82*(1), 137–143. https://doi.org/10.1037/h0033815

Rescorla, R. A., & Solomon, R. L. (1967). Two-process learning theory: Relationships between Pavlovian conditioning and instrumental learning. *Psychological Review*, *74*(3), 151–182. https://doi.org/10.1037/h0024475

Rescorla, R. A., & Wagner, A. R. (1972). A theory of Pavlovian conditioning: Variations in the effectiveness of reinforcement and nonreinforcement. In A. H. Black & W. F. Prokasy (Eds.), *Classical conditioning II: Current research and theory* (pp. 64–99). Appleton Century Crofts.

Revusky, S. H. (1968). Aversion to sucrose produced by contingent x-irradiation: Temporal and dosage parameters. *Journal of Comparative and Physiological Psychology*, *65*(1), 17–22. https://doi.org/10.1037/h0025416

Revusky, S. H. (1971). The role of interference in association over a delay. In W. K. Honig & P. H. R. James (Eds.), *Animal memory* (pp. 155–213). Academic Press. https://doi.org/10.1016/B978-0-12-355050-7.50009-6

Rizley, R. C., & Rescorla, R. A. (1972). Associations in second-order conditioning and sensory preconditioning. *Journal of Comparative and Physiological Psychology*, *81*(1), 1–11. https://doi.org/10.1037/h0033333

Robinson, M. J. F., Anselme, P., Fischer, A. M., & Berridge, K. C. (2014). Initial uncertainty in Pavlovian reward prediction persistently elevates incentive salience and extends sign-tracking to normally unattractive cues. *Behavioural Brain Research*, *266*, 119–130. https://doi.org/10.1016/j.bbr.2014.03.004

Robinson, M. J. F., & Berridge, K. C. (2013). Instant transformation of learned repulsion into motivational "wanting." *Current Biology*, *23*(4), 282–289. https://doi.org/10.1016/j.cub.2013.01.016

Roxburgh, A. D., Hughes, M. E., & Cornwell, B. R. (2019). Threat-induced anxiety weakens inhibitory control. *Biological Psychology*, *144*, 99–102. https://doi.org/10.1016/j.biopsycho.2019.03.009

Salkovskis, P. M., Clark, D. M., Hackmann, A., Wells, A., & Gelder, M. G. (1999). An experimental investigation of the role of safety-seeking behaviours in the maintenance of panic disorder with agoraphobia. *Behaviour Research and Therapy*, *37*(6), 559–574. https://doi.org/10.1016/S0005-7967(98)00153-3

Schell, A. M., Dawson, M. E., & Marinkovic, K. (1991). Effects of potentially phobic conditioned stimuli on retention, reconditioning, and extinction of the conditioned skin conductance response. *Psychophysiology*, *28*(2), 140–153. https://doi.org/10.1111/j.1469-8986.1991.tb00403.x

Scherer, K. R., & Moors, A. (2019). The emotion process: Event appraisal and component differentiation. *Annual Review of Psychology*, *70*(1), 719–745. https://doi.org/10.1146/annurev-psych-122216-011854

Schultz, W., & Romo, R. (1990). Dopamine neurons of the monkey midbrain: Contingencies of responses to stimuli eliciting immediate behavioral reactions. *Journal of Neurophysiology*, *63*(3), 607–624. https://doi.org/10.1152/jn.1990.63.3.607

Seligman, M. E. (1970). On the generality of the laws of learning. *Psychological Review*, *77*(5), 406–418. https://doi.org/10.1037/h0029790

Shanks, D. R. (1985). Forward and backward blocking in human contingency judgement. *The Quarterly Journal of Experimental Psychology B: Comparative and Physiological Psychology*, *37*(1b), 1–21. https://doi.org/10.1080/14640748508402082

Shanks, D. R. (2017). Regressive research: The pitfalls of post hoc data selection in the study of unconscious mental processes. *Psychonomic Bulletin & Review*, *24*(3), 752–775. https://doi.org/10.3758/s13423-016-1170-y

Shanks, D. R., & St. John, M. F. (1994). Characteristics of dissociable human learning systems. *Behavioral and Brain Sciences*, *17*(3), 367–395. https://doi.org/10.1017/S0140525X00035032

Sheffield, F. D. (1965). Relation between classical conditioning and instrumental learning. In W. F. Prokasy (Ed.), *Classical conditioning: A symposium* (pp. 302–322). Appleton-Century-Crofts.

Siegel, S. (1975). Evidence from rats that morphine tolerance is a learned response. *Journal of Comparative and Physiological Psychology*, *89*(5), 498–506. https://doi.org/10.1037/h0077058

Silva, K. M., & Timberlake, W. (1997). A behavior systems view of conditioned states during long and short CS–US intervals. *Learning and Motivation*, *28*(4), 465–490. https://doi.org/10.1006/lmot.1997.0986

Singh, K., Dawson, M. E., Schell, A. M., Courtney, C. G., & Payne, A. F. (2013). Can human autonomic classical conditioning occur without contingency awareness? The critical importance of the trial sequence. *Biological Psychology*, *93*(1), 197–205. https://doi.org/10.1016/j.biopsycho.2013.02.007

Smith, M. C., Coleman, S. R., & Gormezano, I. (1969). Classical conditioning of the rabbit's nictitating membrane response at backward, simultaneous, and forward CS-US intervals. *Journal of Comparative and Physiological Psychology*, *69*(2), 226–231. https://doi.org/10.1037/h0028212

Spence, K. W. (1956). *Behavior theory and conditioning*. Yale University Press. https://doi.org/10.1037/10029-000

Spoormaker, V. I., Blechert, J., Goya-Maldonado, R., Sämann, P. G., Wilhelm, F. H., & Czisch, M. (2012). Additional support for the existence of skin conductance responses at unconditioned stimulus omission. *NeuroImage*, *63*(3), 1404–1407. https://doi.org/10.1016/j.neuroimage.2012.08.050

Sutton, R. S., & Barto, A. G. (1981). Toward a modern theory of adaptive networks: Expectation and prediction. *Psychological Review*, *88*(2), 135–170. https://doi.org/10.1037/0033-295X.88.2.135

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.

Timberlake, W. (1984). An ecological approach to learning. *Learning and Motivation*, *15*(4), 321–333. https://doi.org/10.1016/0023-9690(84)90001-8

Timberlake, W. (1994). Behavior systems, associationism, and Pavlovian conditioning. *Psychonomic Bulletin & Review*, *1*(4), 405–420. https://doi.org/10.3758/BF03210945

Timberlake, W. (2001). Motivated modes in behavior systems. In R. R. Mowrer & S. B. Klein (Eds.), *Handbook of contemporary learning theories* (pp. 155–209). Erlbaum.

Timberlake, W., & Grant, D. L. (1975). Auto-shaping in rats to the presentation of another rat predicting food. *Science*, *190*(4215), 690–692. https://doi.org/10.1126/science.190.4215.690

Timberlake, W., Wahl, G., & King, D. (1982). Stimulus and response contingencies in the misbehavior of rats. *Journal of Experimental Psychology: Animal Behavior Processes*, *8*(1), 62–85. https://doi.org/10.1037/0097-7403.8.1.62

Tolman, E. C. (1933). Sign-gestalt or conditioned reflex? *Psychological Review*, *40*(3), 246–255. https://doi.org/10.1037/h0071472

Trapold, M. A. (1970). Are expectancies based upon different positive reinforcing events discriminably different? *Learning and Motivation*, *1*(2), 129–140. https://doi.org/10.1016/0023-9690(70)90079-2

Trapold, M. A., & Overmier, J. B. (1972). The second learning process in instrumental learning. In A. H. Black & W. F. Prokasy (Eds.), *Classical conditioning II: Current research and theory* (pp. 427–452). Appleton-Century-Crofts.

Urcelay, G. P., Perelmuter, O., & Miller, R. R. (2008). Pavlovian backward conditioned inhibition in humans: Summation and retardation tests. *Behavioural Processes*, *77*(3), 299–305. https://doi.org/10.1016/j.beproc.2007.07.003

Van Hamme, L. J., & Wasserman, E. A. (1994). Cue competition in causality judgments: The role of nonpresentation of compound stimulus elements. *Learning and Motivation*, *25*(2), 127–151. https://doi.org/10.1006/lmot.1994.1008

Vitay, J., & Hamker, F. H. (2014). Timing and expectation of reward: A neuro-computational model of the afferents to the ventral tegmental area. *Frontiers in Neurorobotics*, *8*, Article 4. https://doi.org/10.3389/fnbot.2014.00004

Wagner, A. R. (1981). SOP: A model of automatic processing in animal behavior. In N. E. Spear & R. R. Miller (Eds.), *Information processing in animals: Memory mechanisms* (pp. 5–47). Lawrence Erlbaum. https://doi.org/10.4324/9781315798820

Wagner, A. R., & Brandon, S. E. (1989). Evolution of a structured connectionist model of Pavlovian conditioning (ÆSOP). In S. B. Klein & R. R. Mowrer (Eds.), *Contemporary learning theories: Pavlovian conditioning and the status of traditional learning theory* (pp. 149–189). Erlbaum.

Wagner, A. R., & Rescorla, R. A. (1972). Inhibition in Pavlovian conditioning: Application of a theory. In R. A. Boakes & H. S. Halliday (Eds.), *Inhibition and learning* (pp. 301–336). Academic Press.

Weidemann, G., & Antees, C. (2012). Parallel acquisition of awareness and differential delay eyeblink conditioning. *Learning & Memory*, *19*(5), 201–210. https://doi.org/10.1101/lm.024851.111

Weidemann, G., Best, E., Lee, J. C., & Lovibond, P. F. (2013). The role of contingency awareness in single-cue human eyeblink conditioning. *Learning & Memory*, *20*(7), 363–366. https://doi.org/10.1101/lm.029975.112

Weidemann, G., & Lovibond, P. F. (2016). The role of US recency in the Perruchet effect in eyeblink conditioning. *Biological Psychology*, *119*, 1–10. https://doi.org/10.1016/j.biopsycho.2016.06.007

Weidemann, G., Satkunarajah, M., & Lovibond, P. F. (2016). I think therefore eyeblink: The importance of contingency awareness in conditioning. *Psychological Science*, *27*(4), 467–475. https://doi.org/10.1177/0956797615625973

White, K., & Davey, G. C. (1989). Sensory preconditioning and UCS inflation in human 'fear' conditioning. *Behaviour Research and Therapy*, *27*(2), 161–166. https://doi.org/10.1016/0005-7967(89)90074-0

Wiens, S., Katkin, E. S., & Öhman, A. (2003). Effects of trial order and differential conditioning on acquisition of differential shock expectancy and skin conductance conditioning to masked stimuli. *Psychophysiology*, *40*(6), 989–997. https://doi.org/10.1111/1469-8986.00117

Williams, D. R. (1965). Classical conditioning and incentive motivation. In W. Prokasy (Ed.), *Classical conditioning: A symposium* (pp. 340–357). Appleton-Century-Crofts.

Williams, D. R., & Williams, H. (1969). Auto-maintenance in the pigeon: Sustained pecking despite contingent non-reinforcement. *Journal of the Experimental Analysis of Behavior*, *12*(4), 511–520. https://doi.org/10.1901/jeab.1969.12-511

Windholz, G., & Lamal, P. A. (1986). Pavlov and the concept of association. *Pavlovian Journal of Biological Science*, *21*(1), 12–15. https://doi.org/10.1007/BF02995034

Woodruff-Pak, D. S., & Steinmetz, J. E. (Eds.). (2000). *Eyeblink classical conditioning: Vol. 1. Applications in humans*. Kluwer Academic.

Woods, S. C. (1976). Conditioned hypoglycemia. *Journal of Comparative and Physiological Psychology*, *90*(12), 1164–1168. https://doi.org/10.1037/h0077292

Zeiner, A., & Grings, W. W. (1968). Backward conditioning: A replication with emphasis on conceptualizations by the subject. *Journal of Experimental Psychology*, *76*(2, Pt. 1), 232–235. https://doi.org/10.1037/h0025374

Zimmer-Hart, C. L., & Rescorla, R. A. (1974). Extinction of Pavlovian conditioned inhibition. *Journal of Comparative and Physiological Psychology*, *86*(5), 837–845. https://doi.org/10.1037/h0036412

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